Sensor-informed reduced-order modelling of street canyon flow and overlying atmospheric boundary layer flow

Comprehensive Exam Presentation

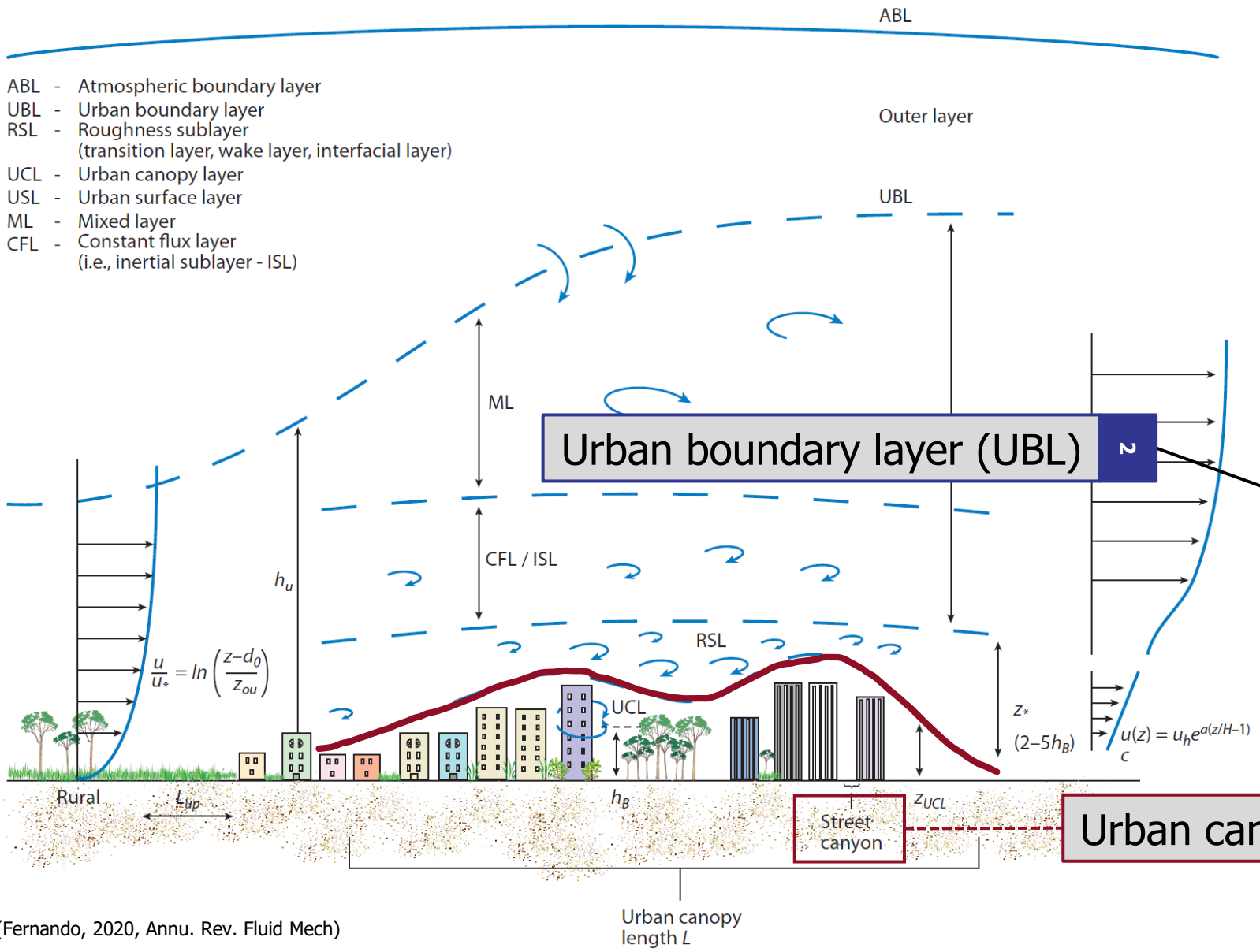
Haoran Du

Supervisor: Eric Savory
Co-supervisor: Christopher DeGroot
International advisor: Laurent Perret

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Motivations & Research Question



(Fernando, 2020, Annu. Rev. Fluid Mech)

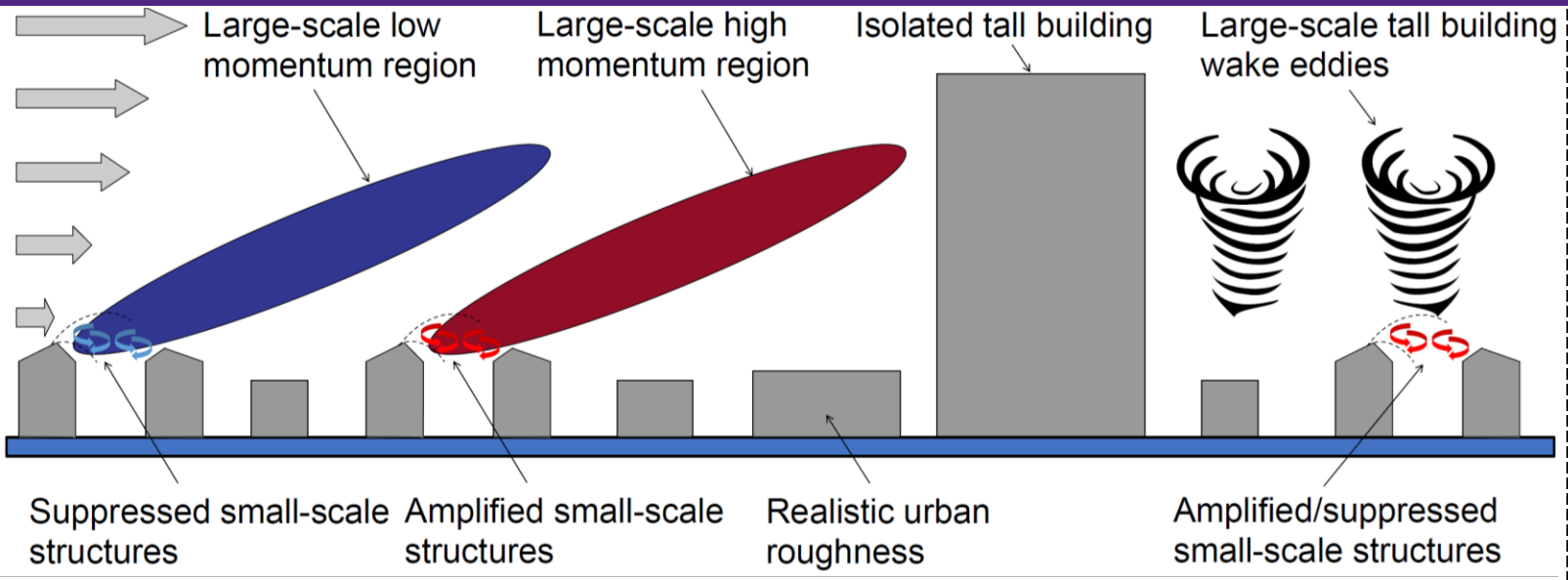


Urban boundary layer (UBL) 2

Dynamic transport of air and pollutants

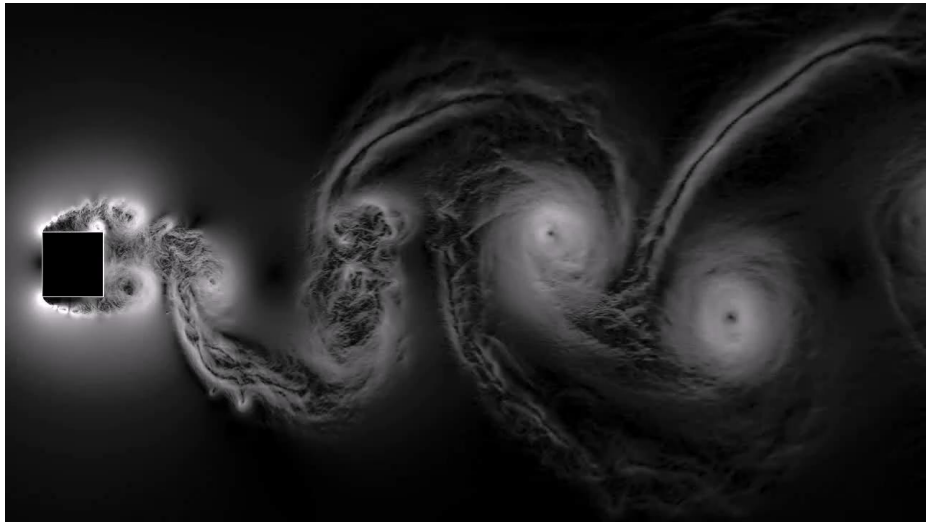
Urban canopy layer (UCL) 1

> Motivation > Background Knowledge



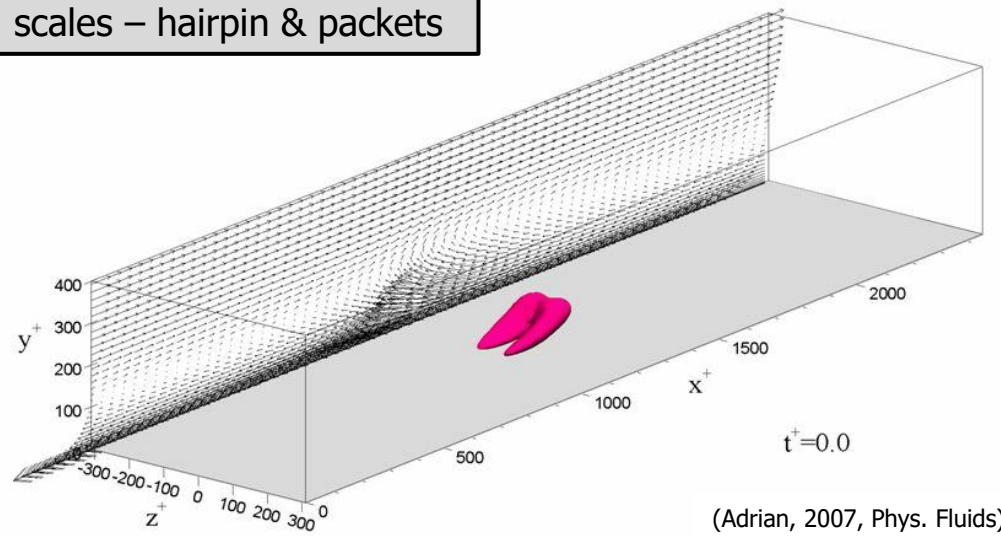
(Du et al., 2023, Bound.-Layer Meteorol.)

Von Kármán vortex street of tall square cylinder



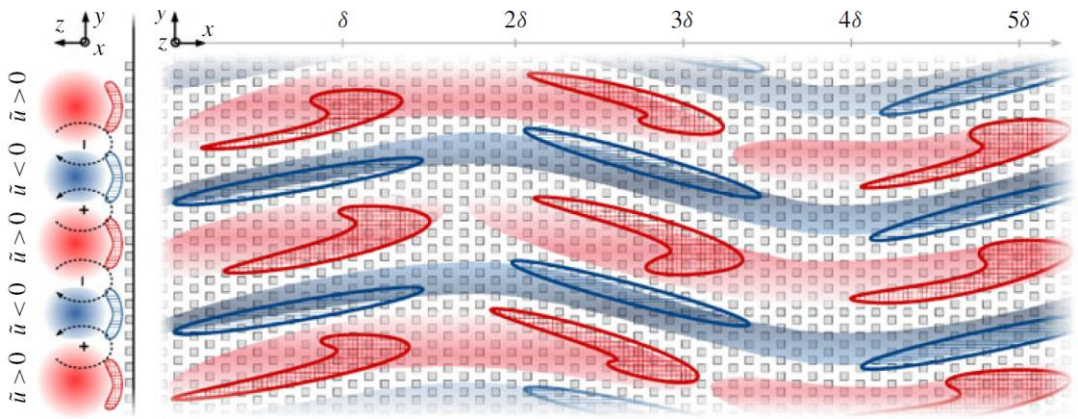
(Trias et al., 2015, Comput. Fluids)

Small scales – hairpin & packets



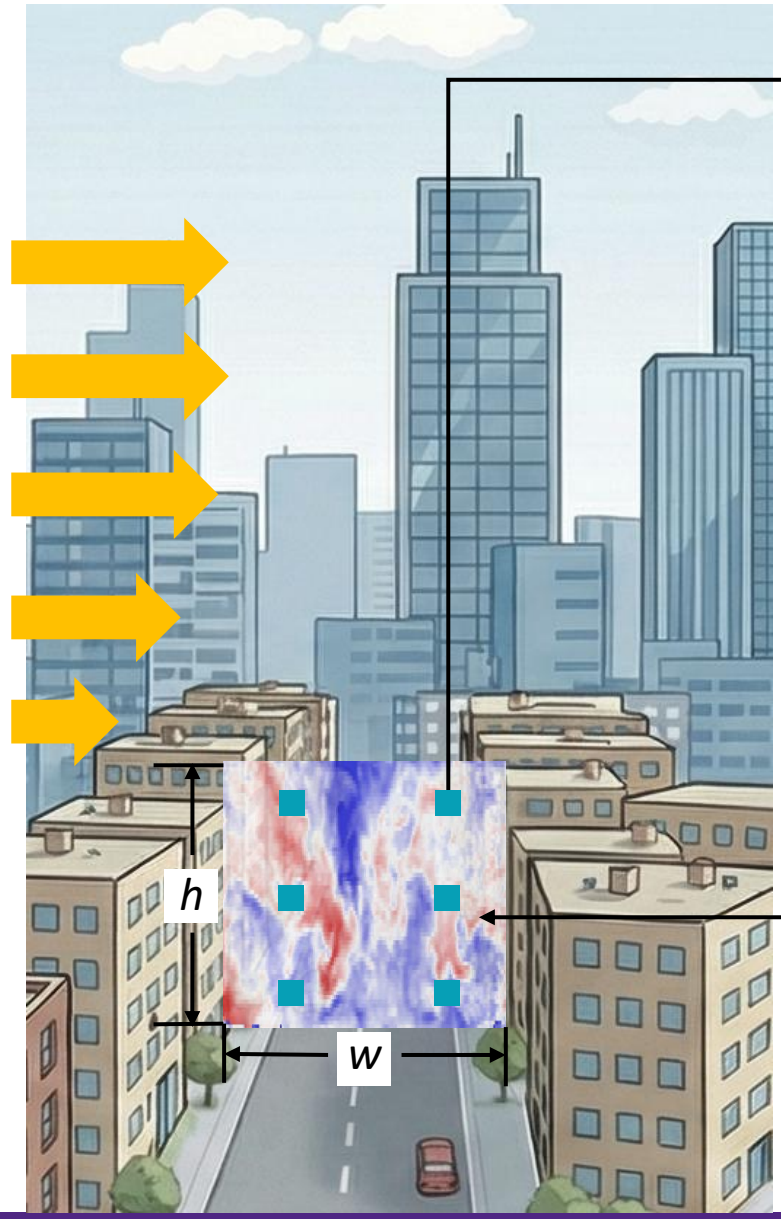
(Adrian, 2007, Phys. Fluids)

Very-large-scale motions (VLSMs)



(Basley et al., 2018, J. Fluid Mech.)

> Research Question



Sparse sensor (time series)

Wind + pollutant measurements

Sensor-informed reduced-order models (ROMs)

POD/SPOD/AE + Mapping (LSE/CNN/RNN)

Reconstructed fields

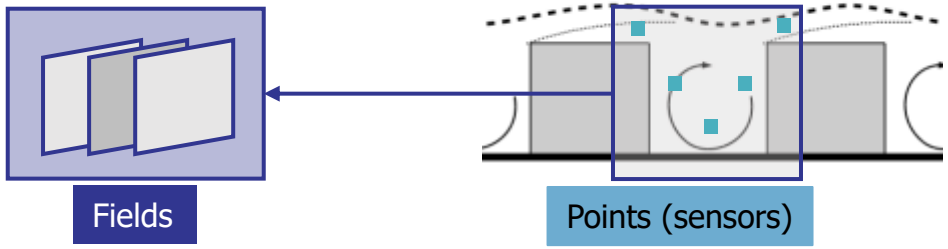
Flow/Scalar maps + ventilation metrics

Research Question
Can sparse time-series measurements **reconstruct** time-resolved urban velocity and pollutant fields **well enough** to quantify ventilation and dispersion across idealised and realistic urban morphologies?
* Definition of "well enough": On held-out cases, the reconstruction matches pointwise error targets and passes physics-based checks (spectra, Reynolds stresses, coherent-structure organisation) and ventilation/dispersion metrics

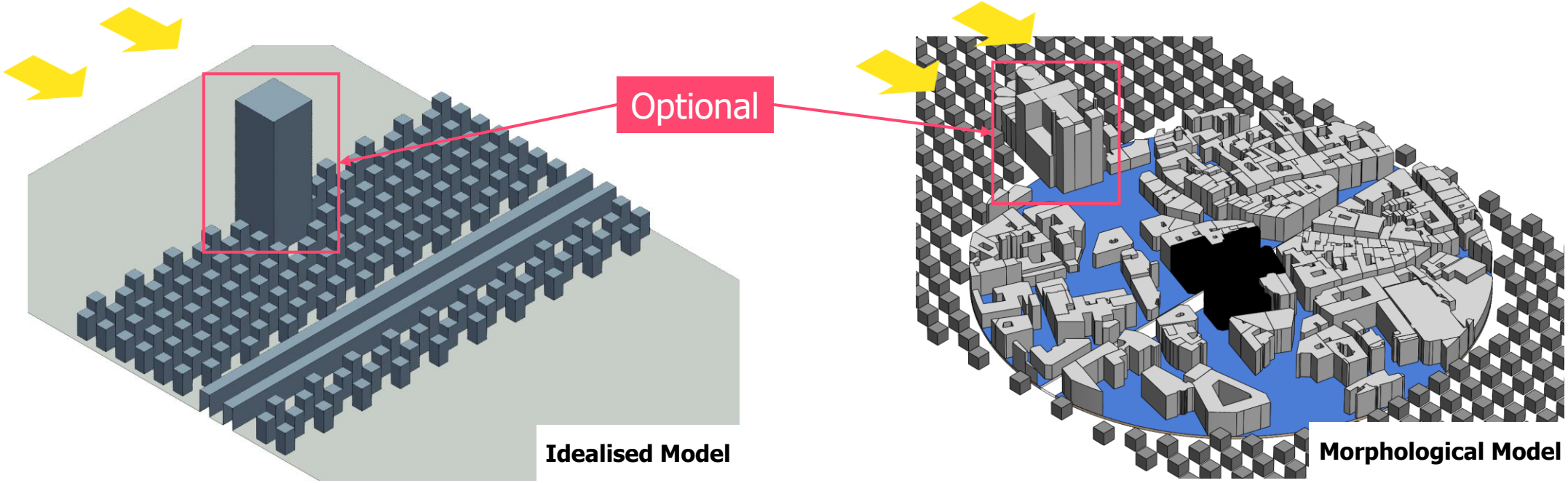
Methods

Real Space (high dimension)

Truth



How to acquire datasets? 🧐



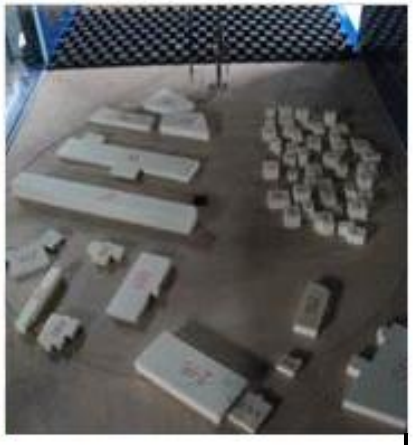
> Methods > Wind tunnel experiments

Atmospheric boundary layer wind tunnel at Ecole Centrale de Nantes

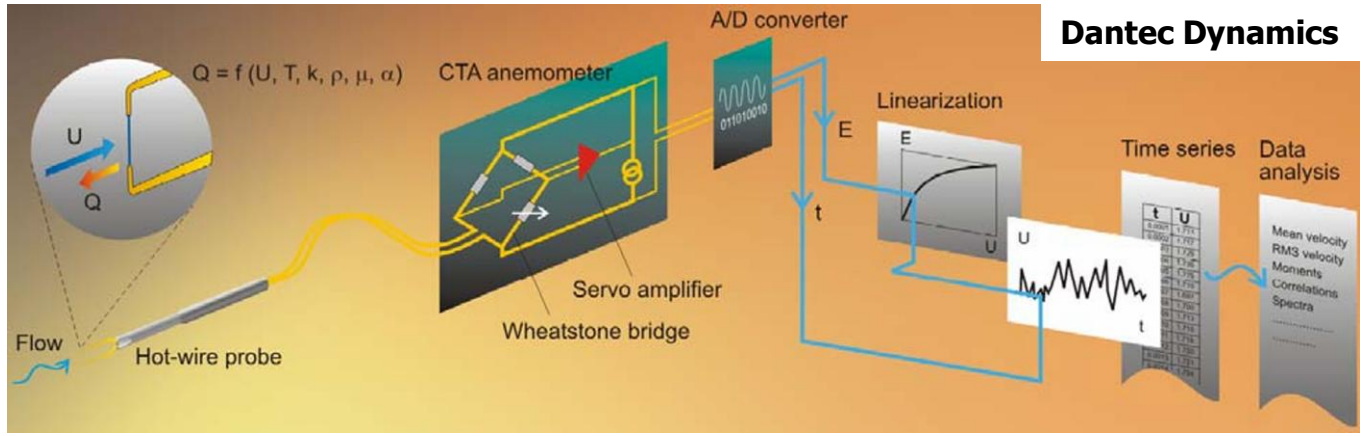




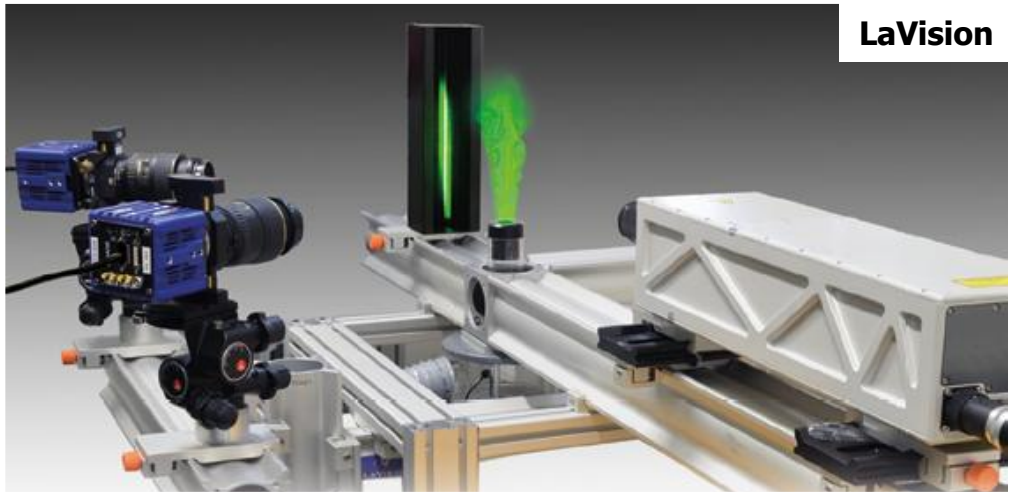




- Controlled inflow & boundary conditions
- Repeatable & well-instrumented
- Isolatable geometry effects

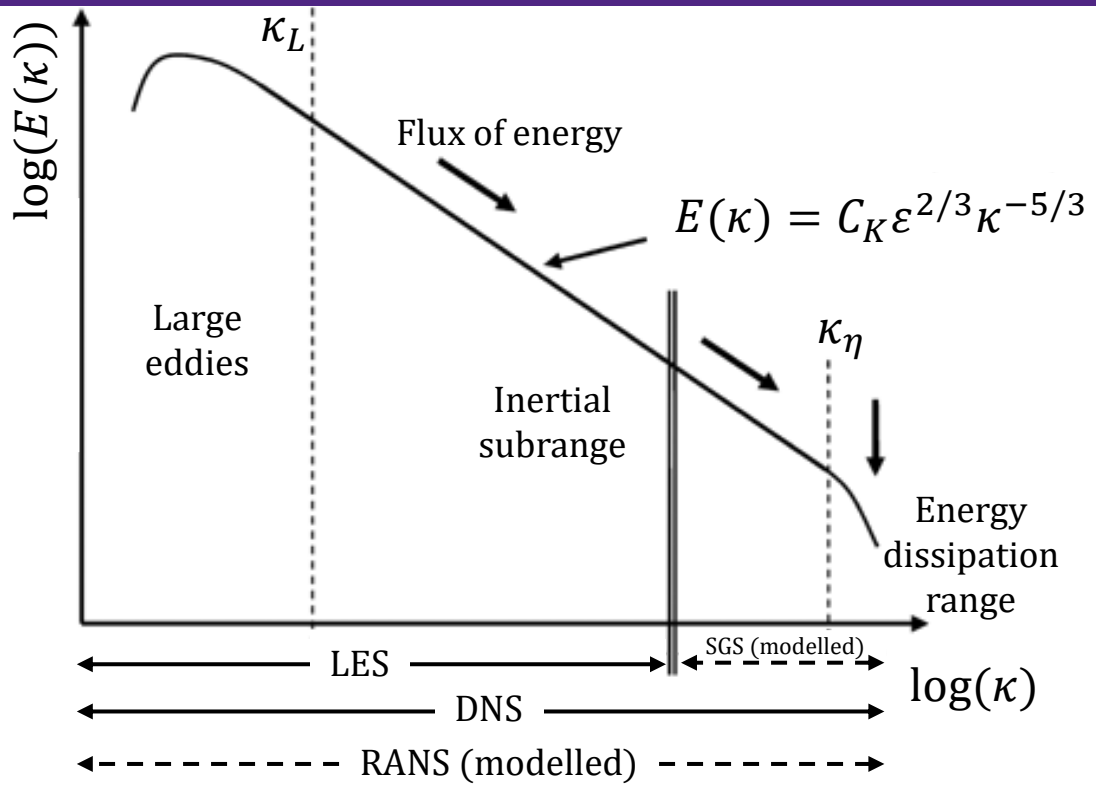


Hot-Wire Anemometry (HWA):
high-frequency time series (sensor-like)

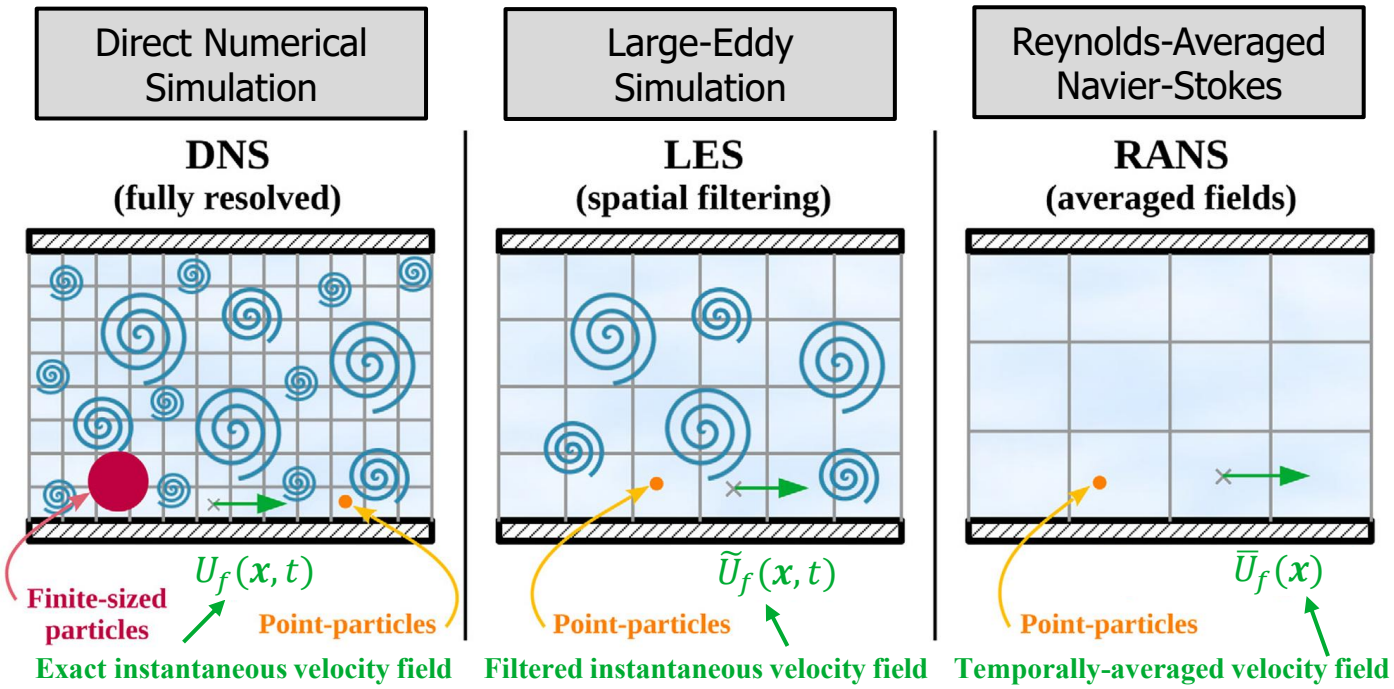


Stereoscopic Particle Image Velocimetry (SPIV) :
planar 3-component (2D3C) velocity fields

> Methods > Numerical Simulations

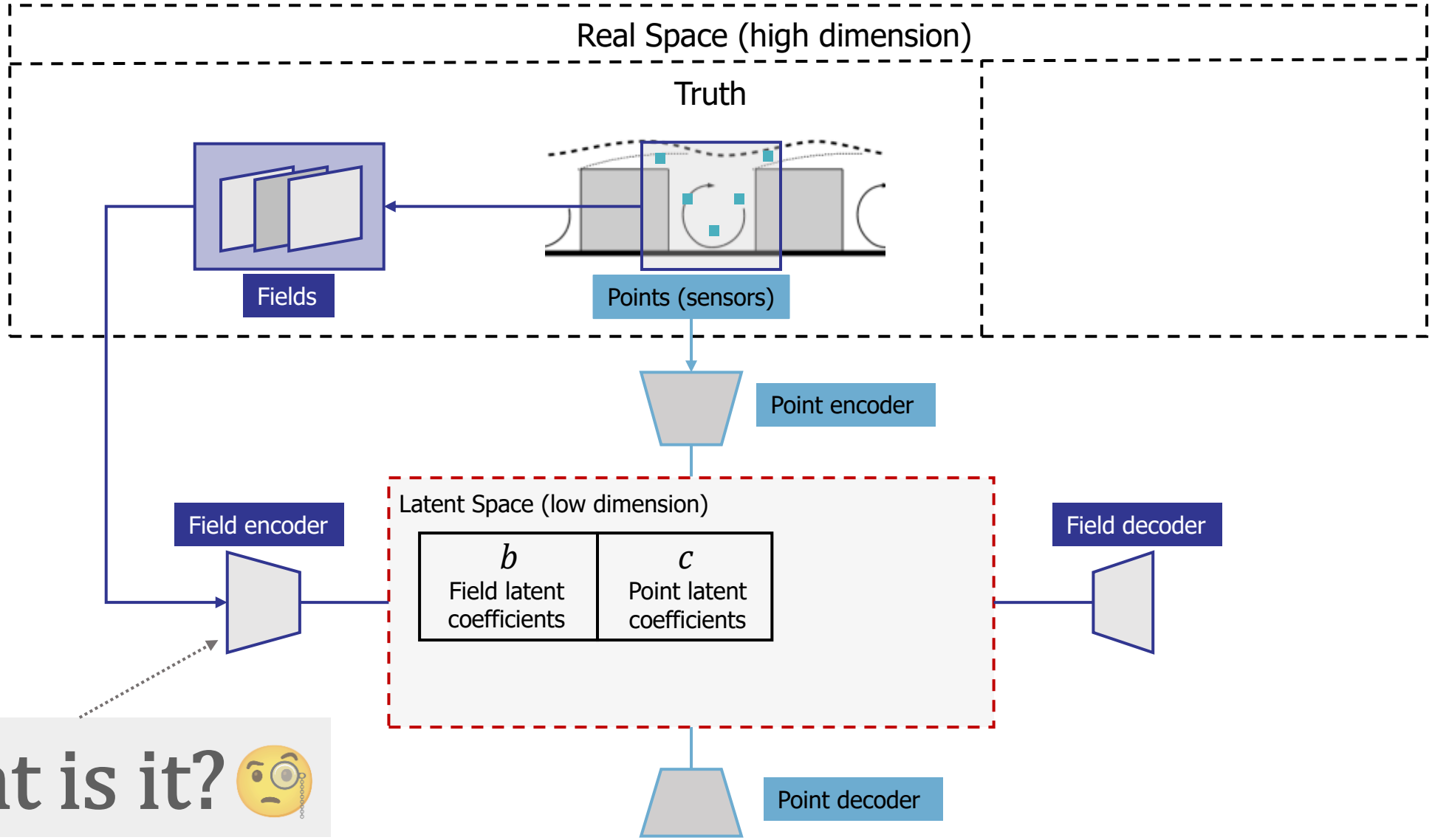


ℓ	characteristic length scale	$\kappa = \frac{2\pi}{\ell}$	wavenumber
L	integral length scale	$\kappa_L = \frac{2\pi}{L}$	integral wavenumber
η	Kolmogorov length scale	$\kappa_\eta = \frac{2\pi}{\eta}$	Kolmogorov wavenumber
Δ	LES filter width	$\kappa_c = \frac{\pi}{\Delta}$	LES cutoff wavenumber
ε	TKE dissipation rate per unit mass		



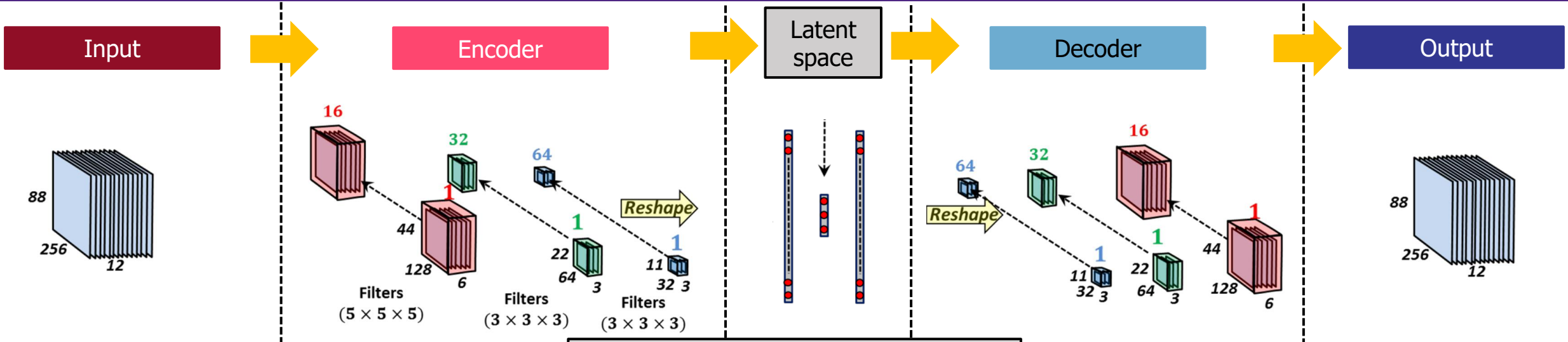
(Henry et al., 2023, Phys. Rep.)

- LES: resolves $\kappa < \kappa_c$ and models sub-grid scales (SGS) for $\kappa > \kappa_c$.
- For fixed Re: finer mesh ⇒ larger κ_c ⇒ a larger fraction of the spectrum is resolved.
- DNS: resolves down to the Kolmogorov scale.
- RANS: solves mean fields; turbulence effects at all scales are represented by a closure model.


Real Space (high dimension)
Truth
Fields
Points (sensors)
Point encoder
Latent Space (low dimension)
Field encoder
b
Field latent coefficients
c
Point latent coefficients
Field decoder
Point decoder

What is it? 🧐

> Methods > Reduced-order representations



(Agostini, 2020, Phys. Fluids)

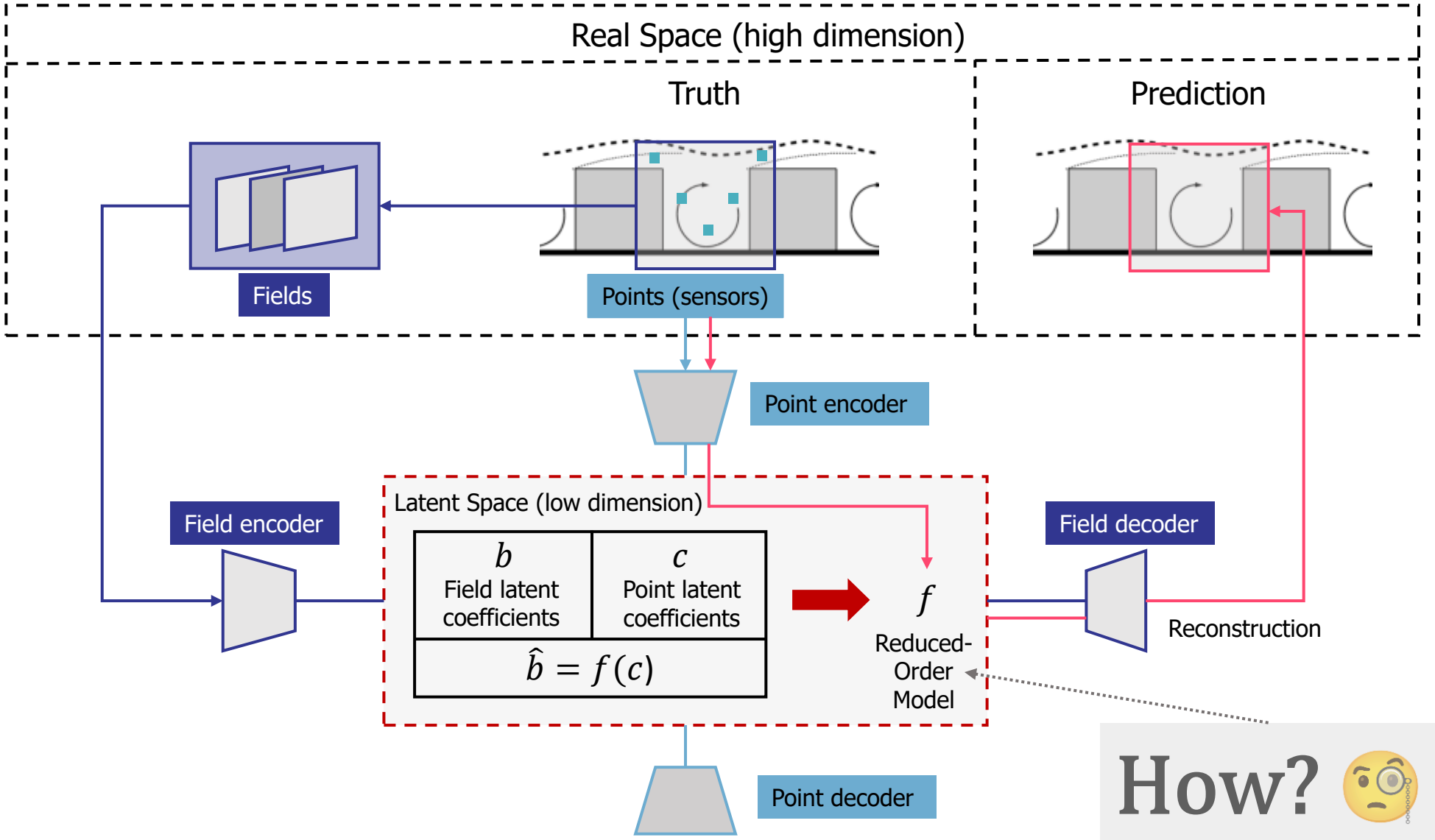
Convolutional Autoencoder (convAE)

x	$g_\theta(x)$	z	$h_\psi(z)$	$\hat{x}$

Proper Orthogonal Decomposition (POD)

Demean & Vectorise each snapshot field: $x \in \mathbb{R}^N$	$z = W_e x$, where $W_e = \mathbf{\Phi}_r^T \in \mathbb{R}^{r \times N}$ $\mathbf{\Phi}_r = [\Phi_1, \dots, \Phi_r]$ orthogonal modes	z	$\hat{x} = W_d z$, where $W_d = \mathbf{\Phi}_r \in \mathbb{R}^{N \times r}$	$\hat{x}$

- POD/SPOD: + interpretable, energy/frequency ranked, − linear only, may need many modes for nonlinear flow.
- AE: + nonlinear, compact latent for strong reconstruction; − less interpretable, not ranked, needs physics validation.


Real Space (high dimension)
Truth
Prediction
Fields
Points (sensors)
Point encoder
Field encoder
Latent Space (low dimension)
b
Field latent coefficients
c
Point latent coefficients
$\hat{b} = f(c)$
f
Reduced-Order Model
Field decoder
Reconstruction
Point decoder
How? 🧐

> Methods > Reduced-Order Modelling (Mapping)

- LSE: + fast, interpretable baseline; – linear only (misses intermittent/nonlinear events).
- QSE: + adds quadratic nonlinearity; – many parameters → data-hungry/overfit risk.
- CNN: + strong nonlinear field reconstruction; – needs lots of data, sensitive to domain shift.
- RNN: + captures time memory; – harder to train, sensitive to domain shift.

Linear Stochastic Estimation (LSE)

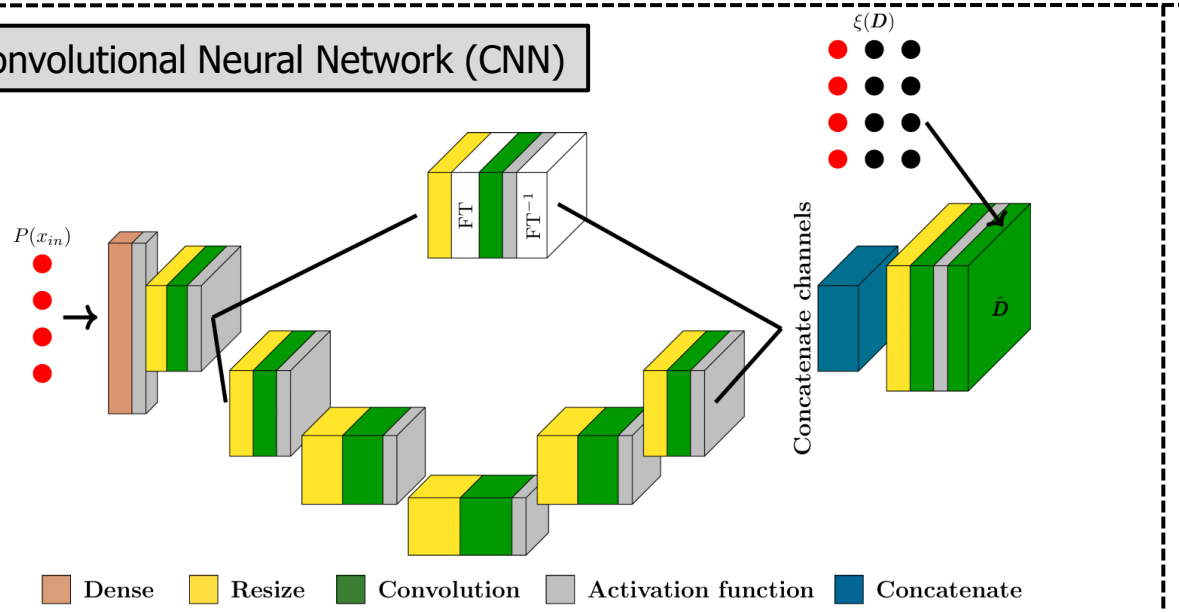
$$\tilde{u}_i(x,z,t) = \sum_{j=1}^{N_c} \sum_{l=1}^{N_{ref}} A^i_{jl}(x,z) u_j(x_l, z_l, t)$$

Quadratic Stochastic Estimation (QSE)

$$\tilde{u}_i(x,z,t) = \sum_{j=1}^{N_c} \sum_{l=1}^{N_{ref}} A^i_{jl}(x,z) u_j(x_l, z_l, t) + \sum_{j,l=1}^{N_c} \sum_{m,n=1}^{N_{ref}} B^i_{jlmn}(x,z) u_j(x_m, z_m, t) u_j(x_n, z_n, t)$$

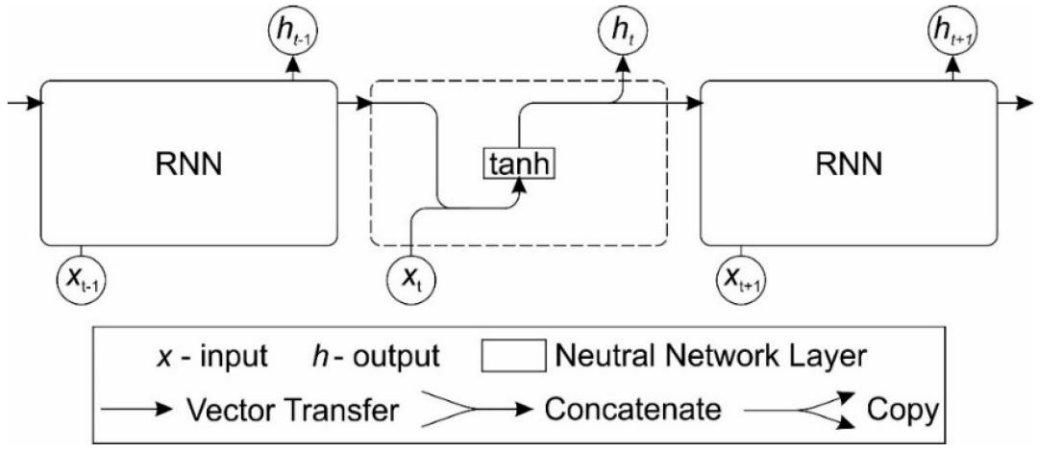
(Perret et al., 2023, Bound.-Layer Meteorol.)

Convolutional Neural Network (CNN)



(Mo & Magri, 2025, Phys. Rev. Fluids)

Recurrent Neural Network (RNN)



(Skrobek et al. 2020, Energies)

Testable Objectives

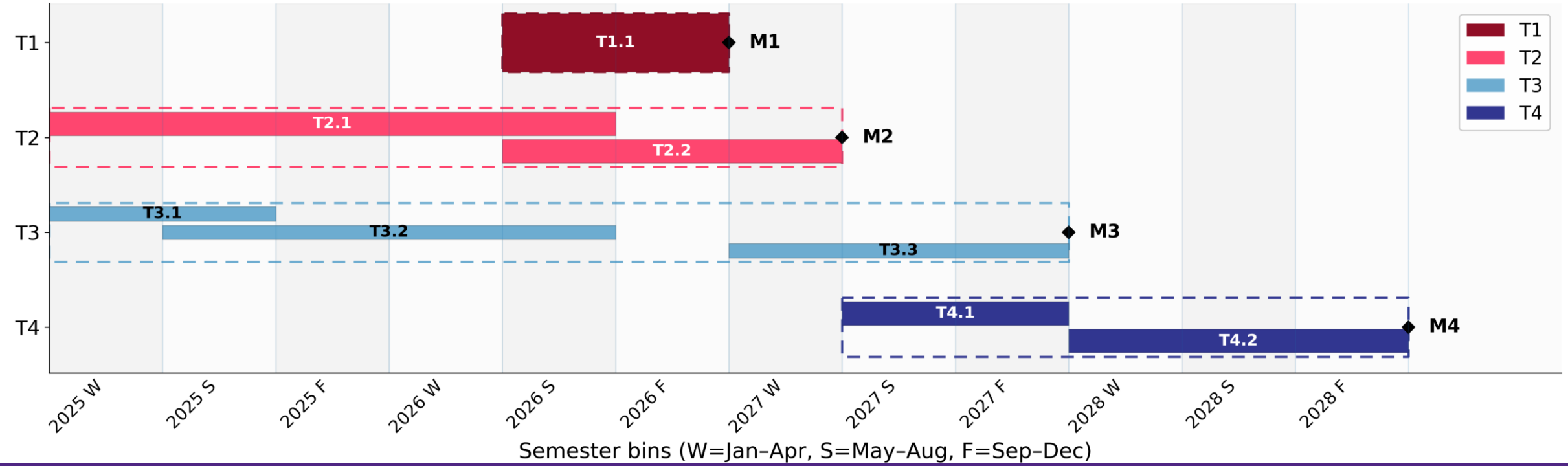
> Testable objectives

O1: Reduced-order representations (ABL and UCL flow)
Build POD/SPOD/AE reduced representations for velocity (and passive scalar) and quantify reconstruction error vs rank/latent size on LES and SPIV planes.
O2: Sensor-to-field mapping (ABL and UCL flow)
With fixed sensor layout, train and compare LSE/CNN/RNN, evaluate on held-out cases with pointwise error + physics checks.
O3: Generalisation under dataset changes
Test transfer from idealised to morphological cases, and from LES to wind tunnel, and report performance drop.
O4: Ventilation/dispersion metrics
Quantify how well reconstructed fields reproduce ventilation/dispersion indicators (roof exchange, pedestrian-level concentration statistics, etc.).
O5: Robust & practical sensor placement
Identify optimal sensor layouts that minimise metric error and still remain robust to noise and dataset changes.

Timeline

> Timeline > Gantt Chart & Milestones

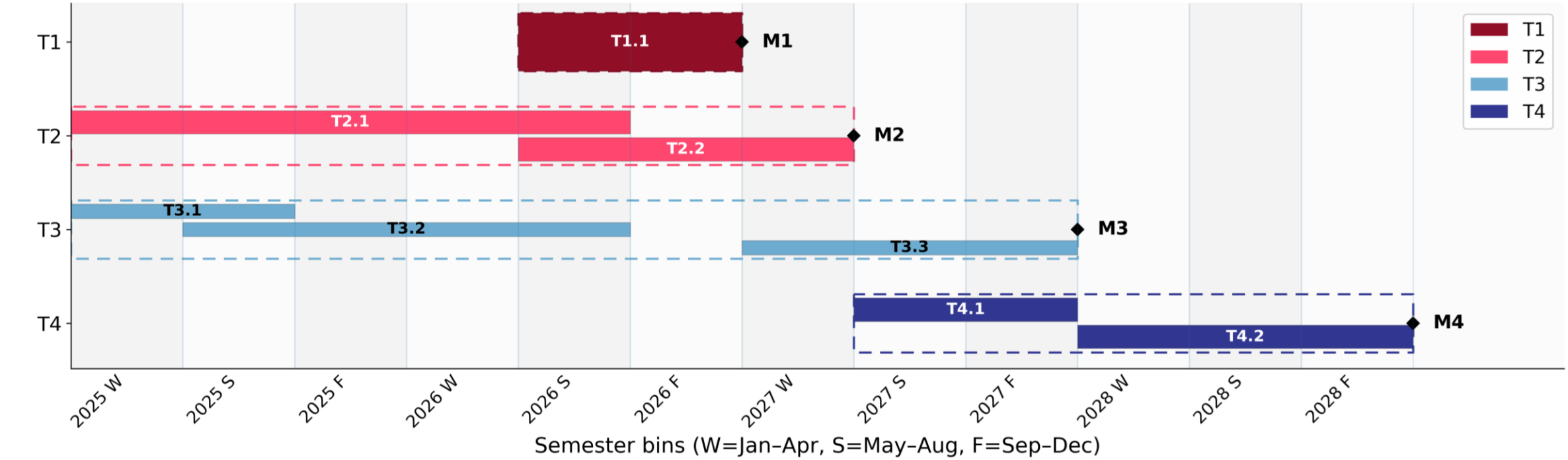
T1.1: Wind tunnel experiments (idealised b1/2/3/6h WT cases)	M1: Idealised wind tunnel dataset analysed + archived
T2.1: Precursor ABL LES simulation T2.2: Successor LES simulations (idealised b1/2/3/6h LES cases)	M2: Validated precursor-successor LES baseline + dispersion capability
T3.1: SPOD/AE for ABL T3.2: Sensor-informed ROM for ABL reconstruction (RdS WT) T3.3: Validate this ROM in idealised WT/LES datasets.	M3: Reduced-order coupling analysis & ABL reconstruction
T4.1: SPOD/AE & ROM for street canyon flow (idealised LES) T4.2: Validate this ROM in idealised WT or RdS WT datasets.	M4: Robust street canyon flow reconstruction + optimal sensor placement guidelines



Current Progress

> Current Progress (Summarised)

- Morphological wind tunnel dataset (RdS) was acquired in my MESc (Published as Du et al. 2023, Build. Environ).
- Idealised wind tunnel experiment is planned and will be carried out in Jun.-Aug. 2026. **(T1.1)**
- A fully-developed ABL is modelled using LES as a precursor (Reported in CFDSC2025). **(T2.1)**
- The successor simulation model and structured mesh are generated. **(part of T2.2)**
- Reduced-order representations (SPOD/AE) for ABL is explored (Published as Du et al. 2025, Bound.-Layer Meteorol). **(T3.1)**
- LSE/CNN are explored for large-scale ABL flow reconstruction with sensor (HWA) informed using RdS dataset (To be submitted to Exp. Fluids in Mar. 2026). **(T3.2)**
- CNN is used for demo sensor-informed ROM for street canyon flow (RdS). It will be updated using idealised LES datasets in the future. **(part of T4.1)**



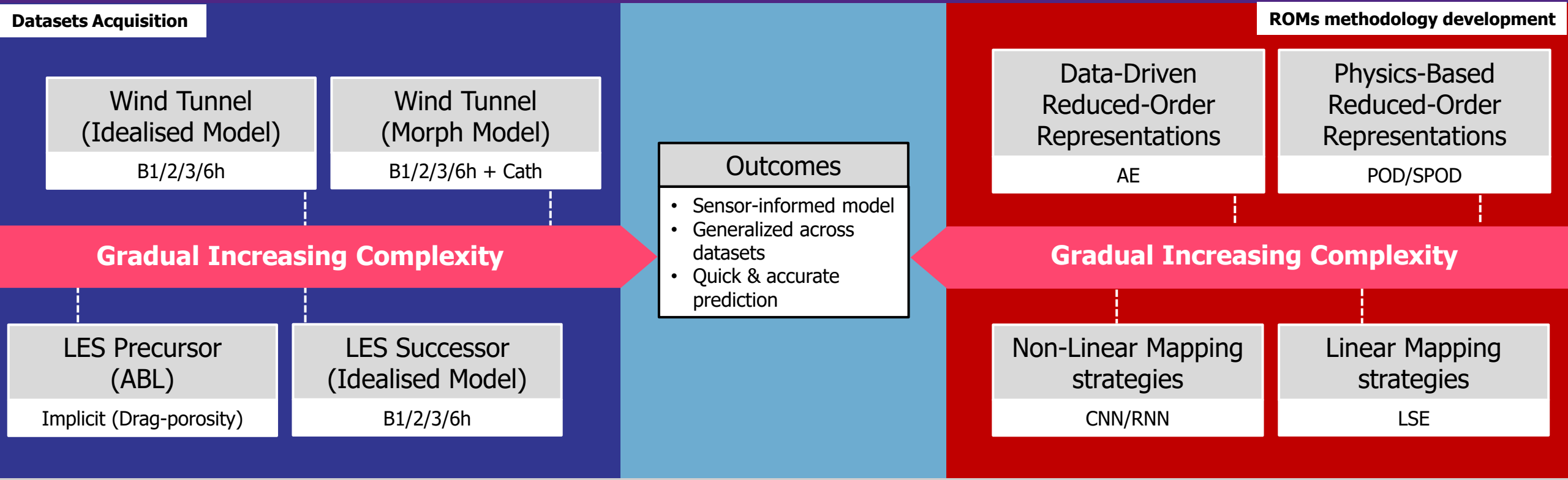
Thank you.

Q & A

- **Current Progress & Future work (detailed)**

  Only upon request in Q&A

> Risk Management



LES sensitivity (successor alignment/BC/mesh): credibility

Mitigation: Grid independence test, validation with idealised WT

Observability limit (sparse + planar): ambiguity in three-dimensional (3D) flow

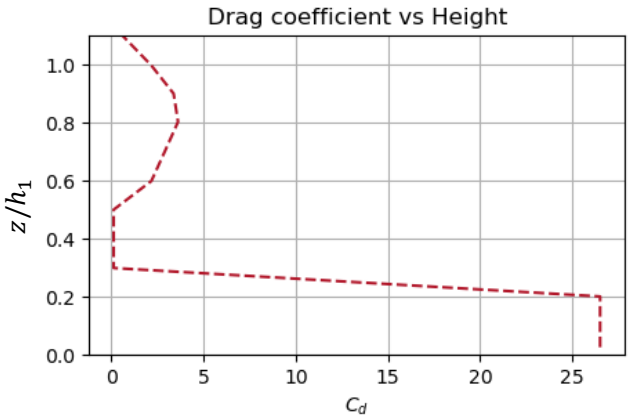
Mitigation: LES 3D sensor layout design + validation on idealised/morph WT planes

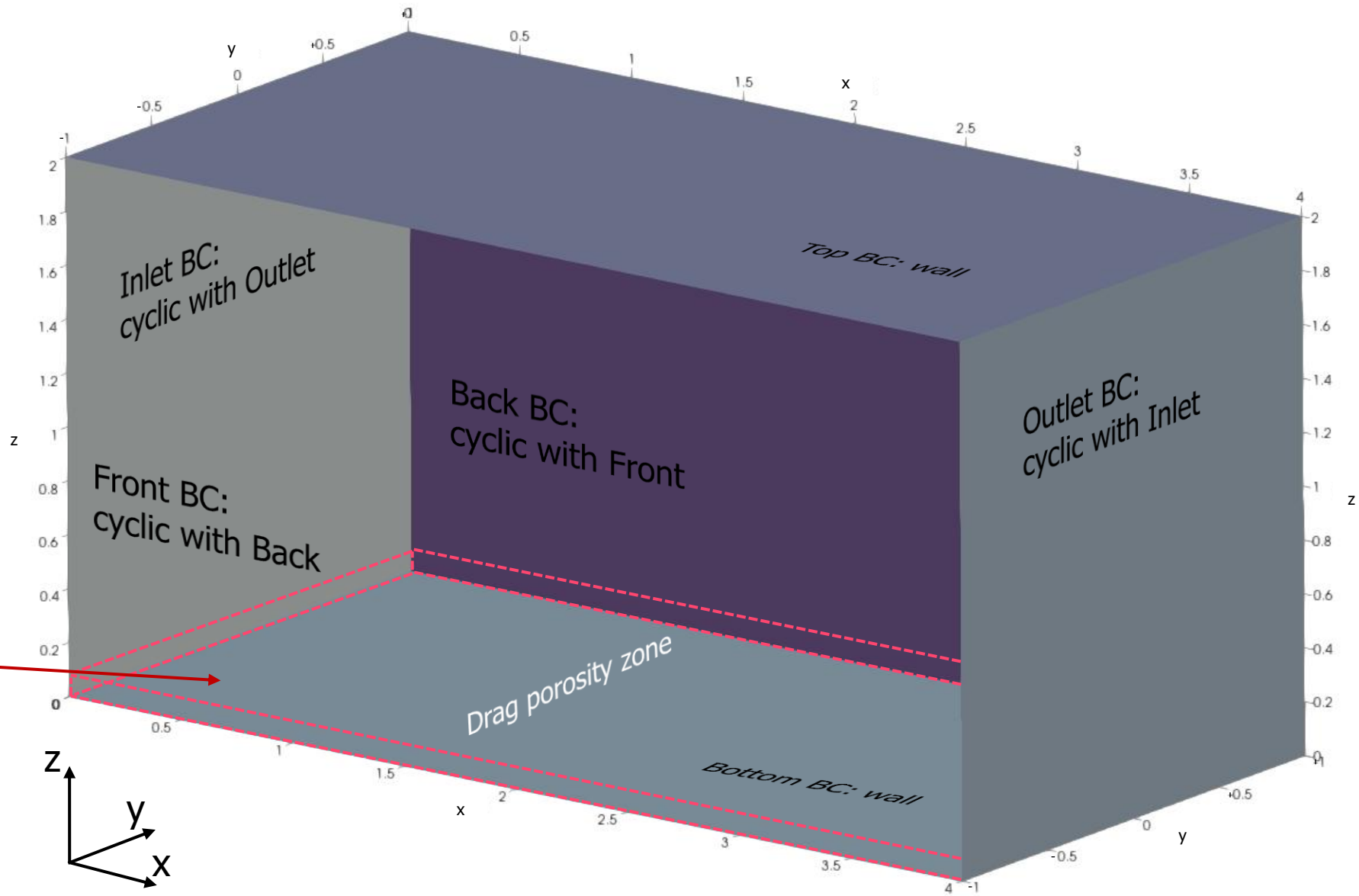
Generalisation risk (distribution shift): idealised↔realistic, WT↔LES

Mitigation: nondimensional sensor layouts, fine-tuning, failure scenios

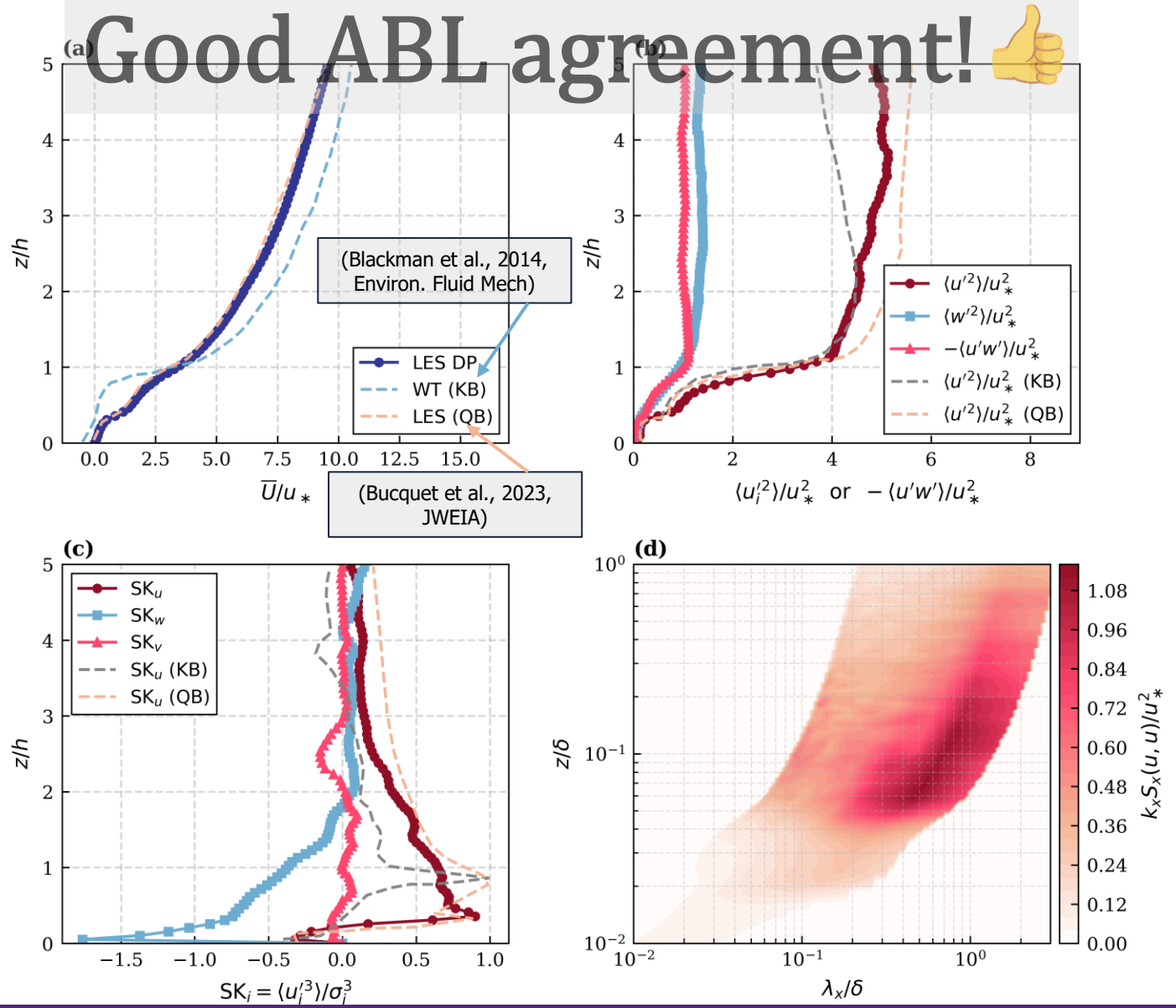
Precursor: Fully developed ABL

- OpenFOAM V2412
- Parallel running on Sharcnet
- Pimplefoam LES - dynamicKEqn
- Pressure-gradient driven
- Drag porosity zone height h_1
- Output: Fields U, k at Outlet





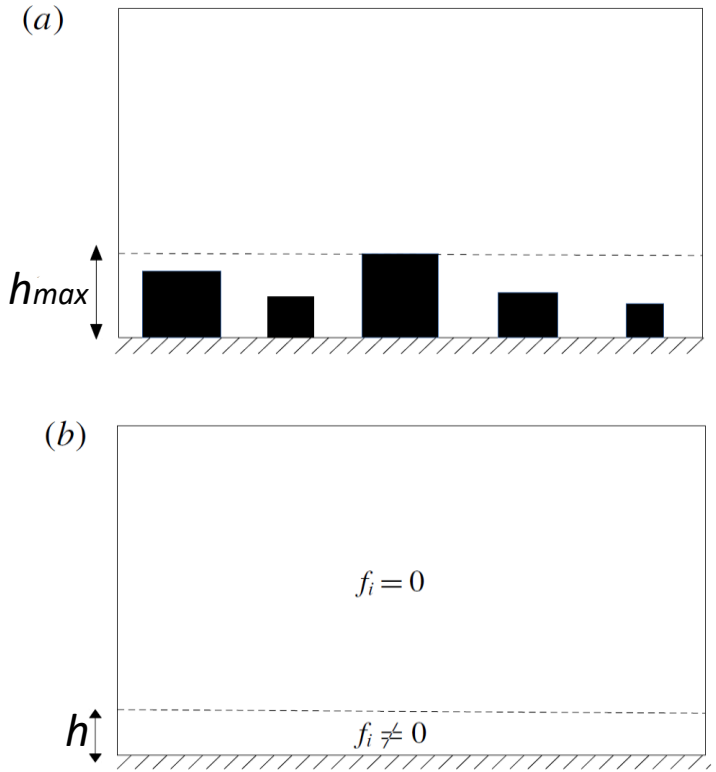
Good ABL agreement!



Drag porosity method

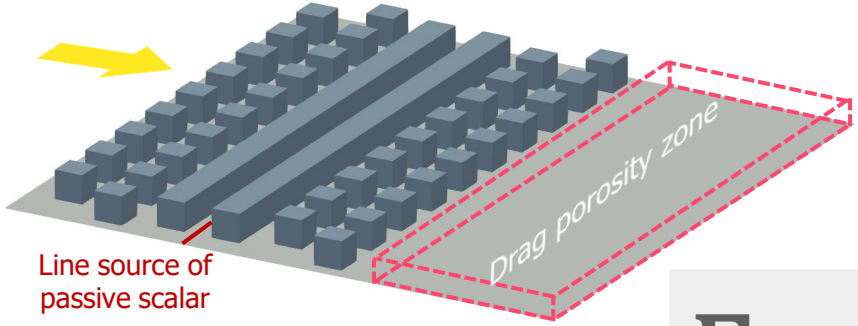
$$\frac{\partial \hat{u}_i}{\partial t} + \hat{u}_j \frac{\partial \hat{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \hat{p}}{\partial x_j} + \frac{\partial}{\partial x_j}\left(\nu \frac{\partial \hat{u}_i}{\partial x_j} - \tau_{ij}^{SGS}\right) - f_i$$

Additional momentum sink

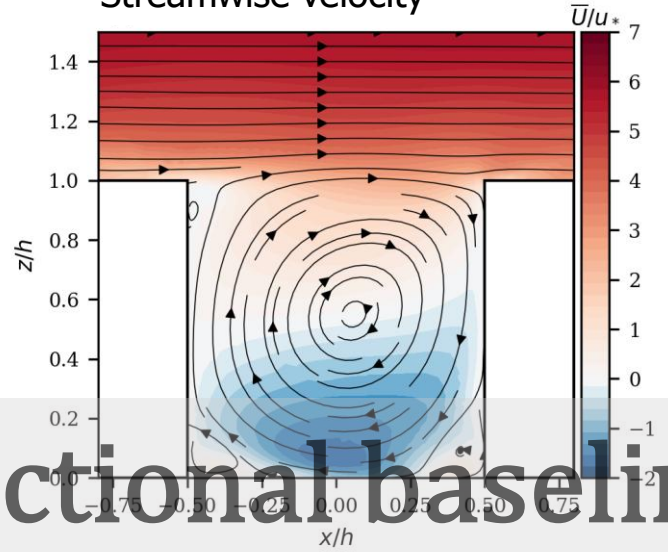


(Varghese and Durbin, 2020, J. Fluid Mech.)

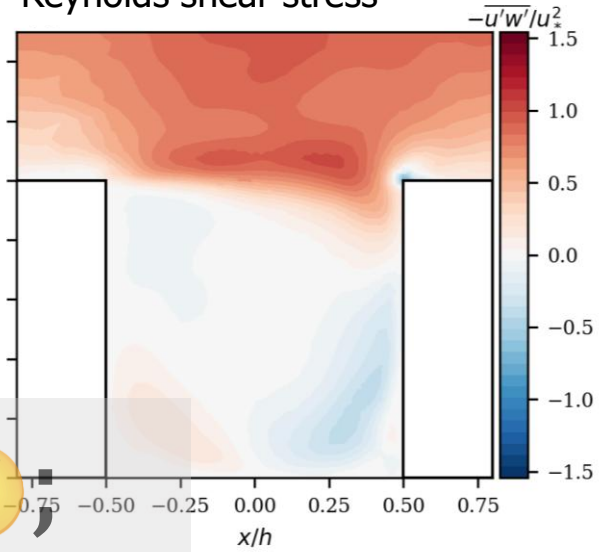
Successor: Preliminary exploration



Streamwise velocity

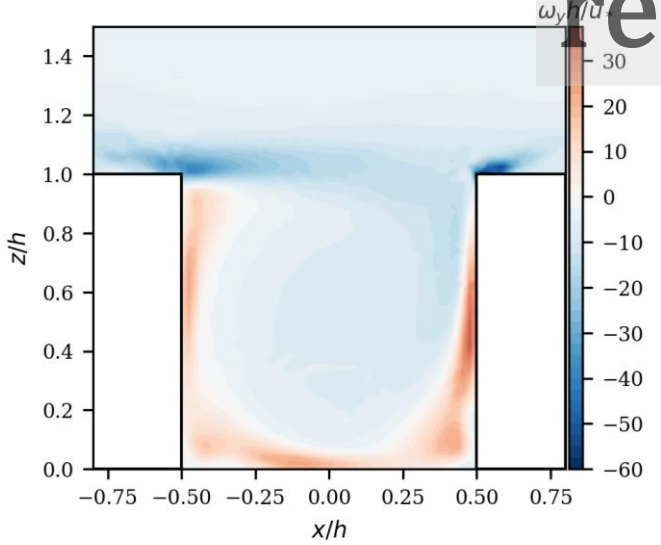


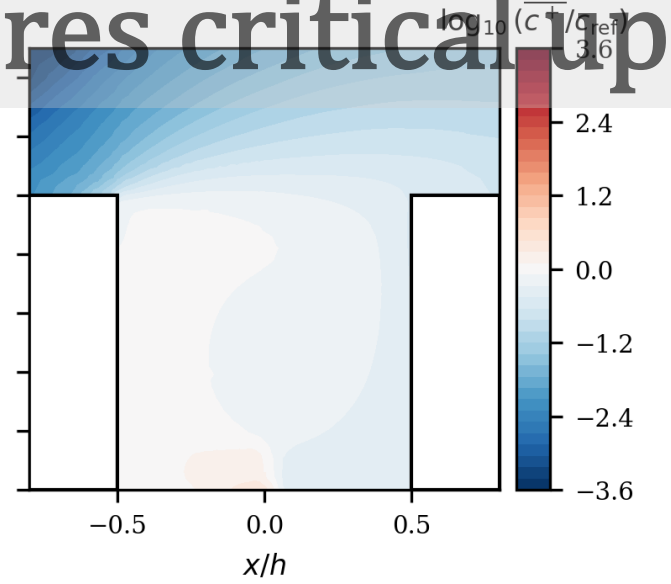
Reynolds shear stress

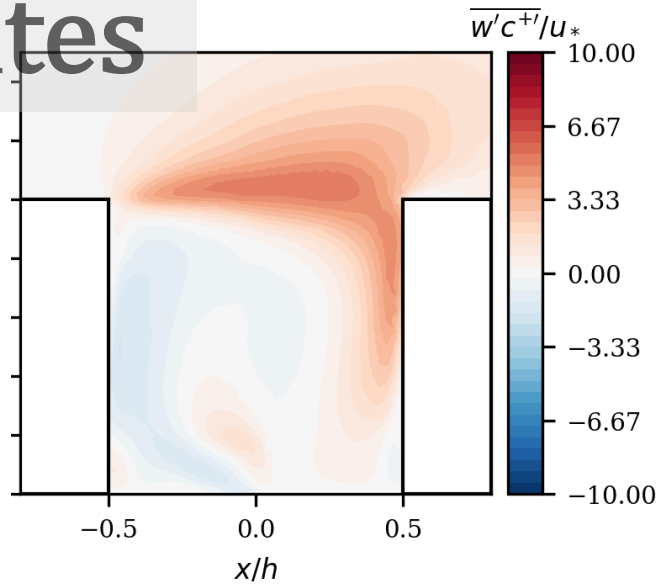


Functional baseline 🤔; requires critical updates

Vorticity

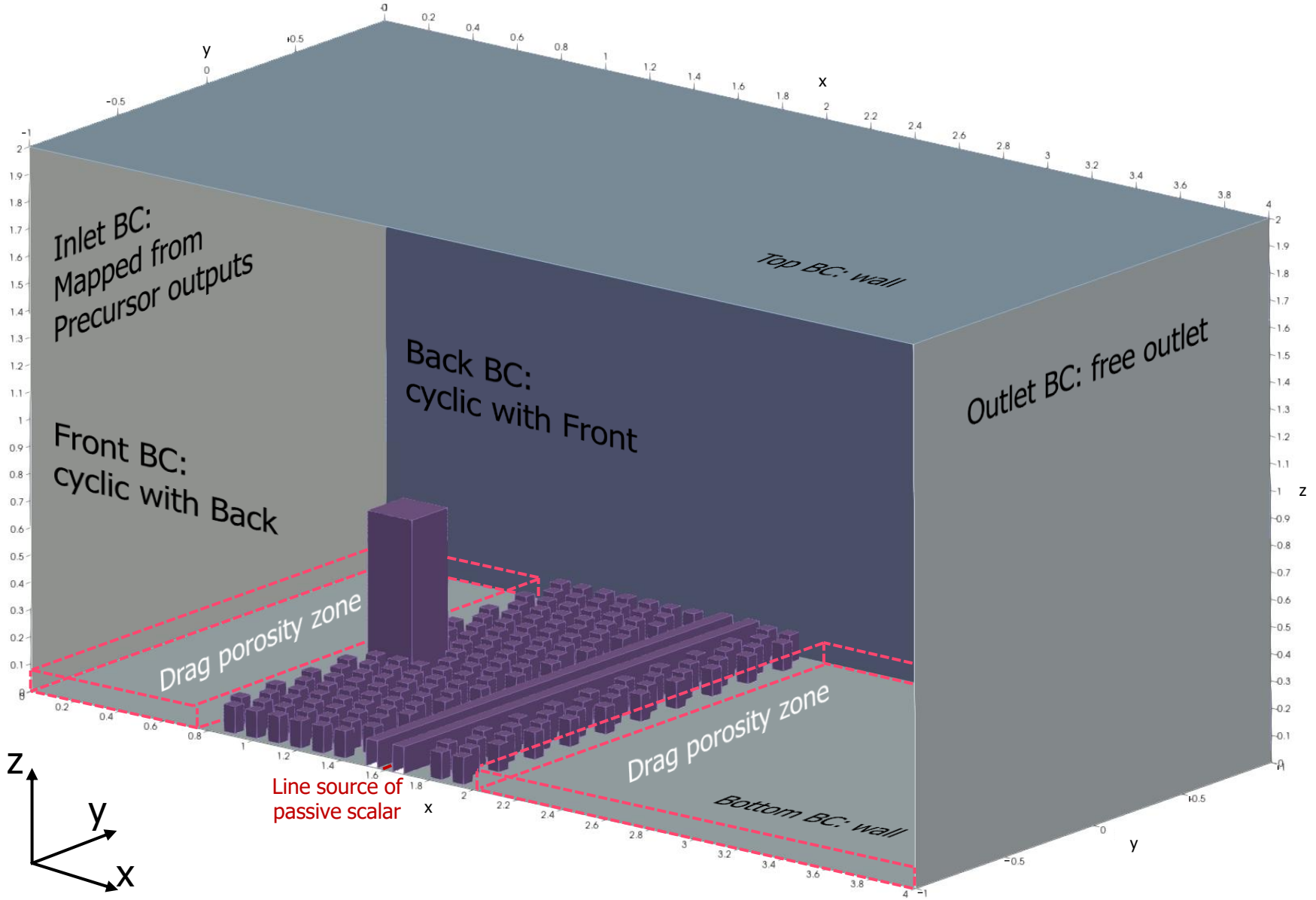




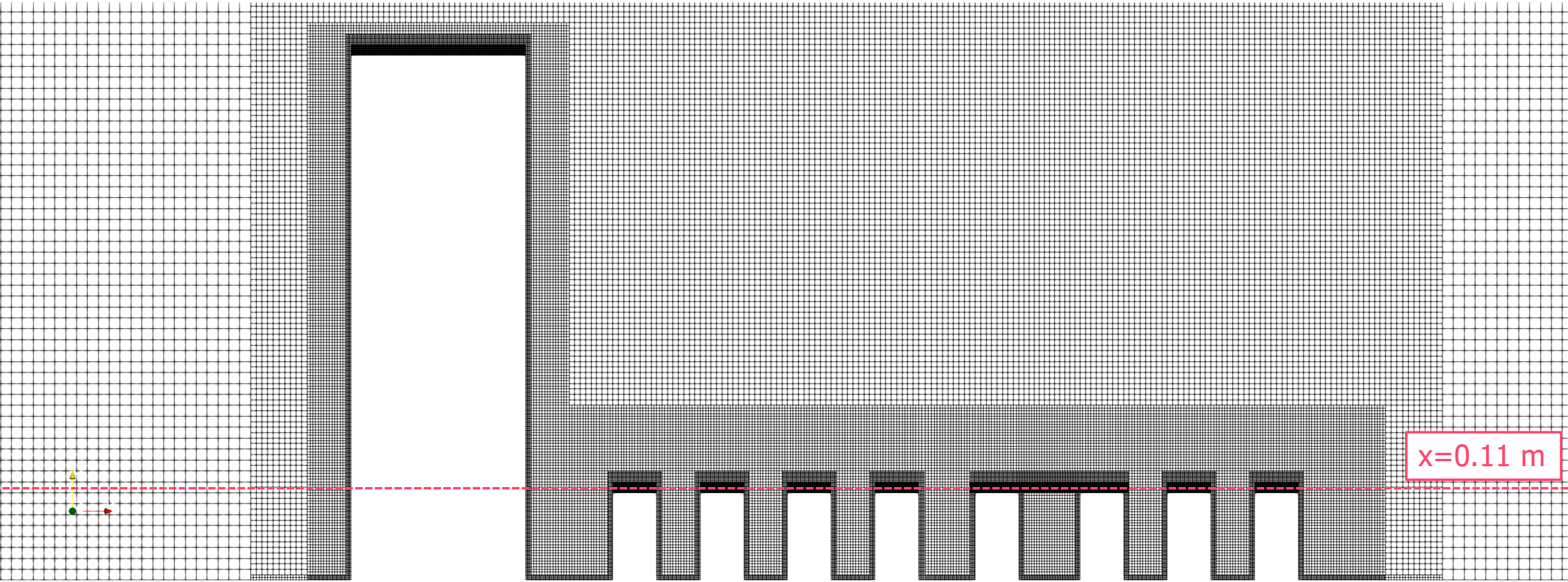


Successor: Target configuration

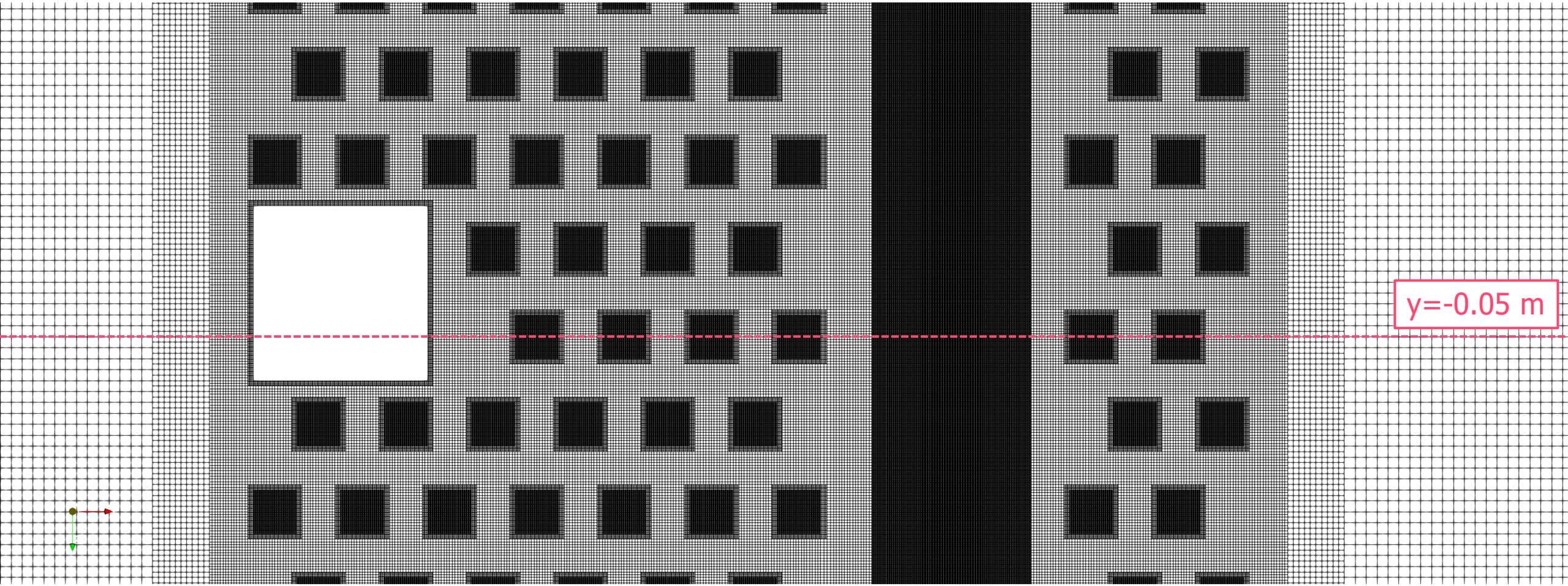
- OpenFOAM V2412
- Parallel running on Sharcnet
- Pimplefoam LES – dynamicKEqn
- Various tall building height
  - 1h/2h/3h/6h (shown)
- Wind tunnel experiments matched
- Passive-scalar line source at canyon floor centre (along y)



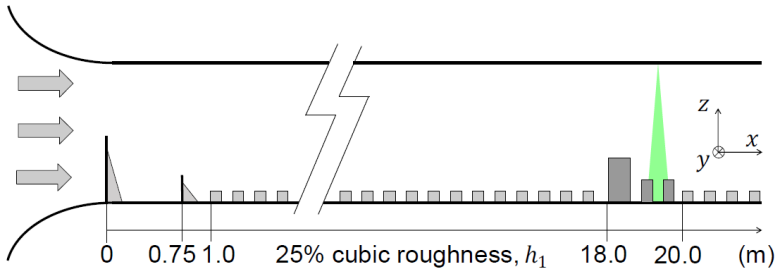
Cross-plane xz (streamwise-vertical) view with mesh gridlines at y=-0.05 m



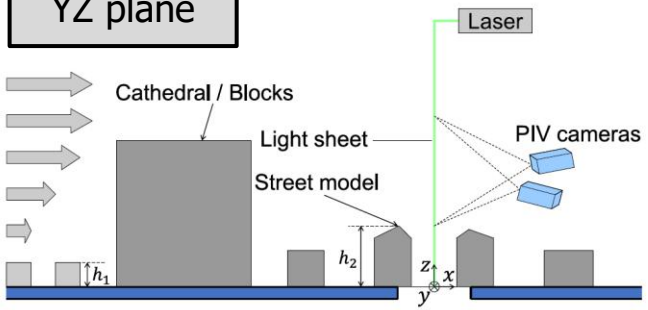
Cross-plane xy (streamwise-transverse) view with mesh gridlines at z=0.11 m

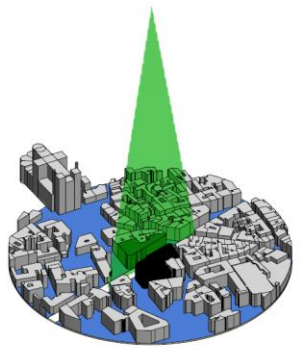


Morphological model Rue de Strasbourg (RdS) dataset – Collected during MESc

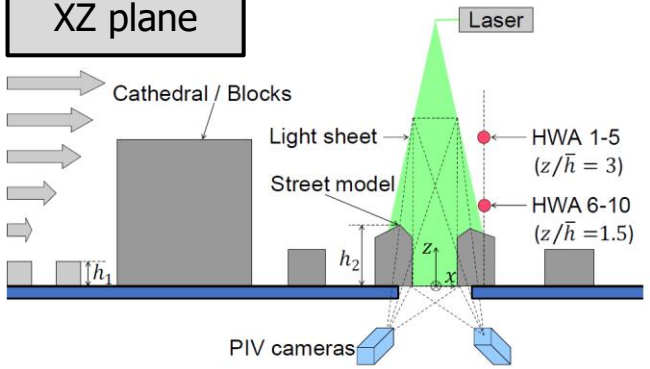


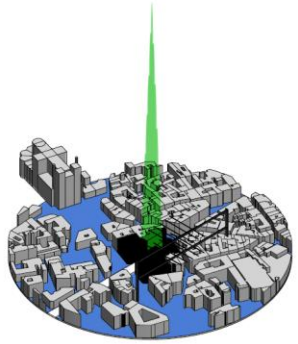
YZ plane





XZ plane





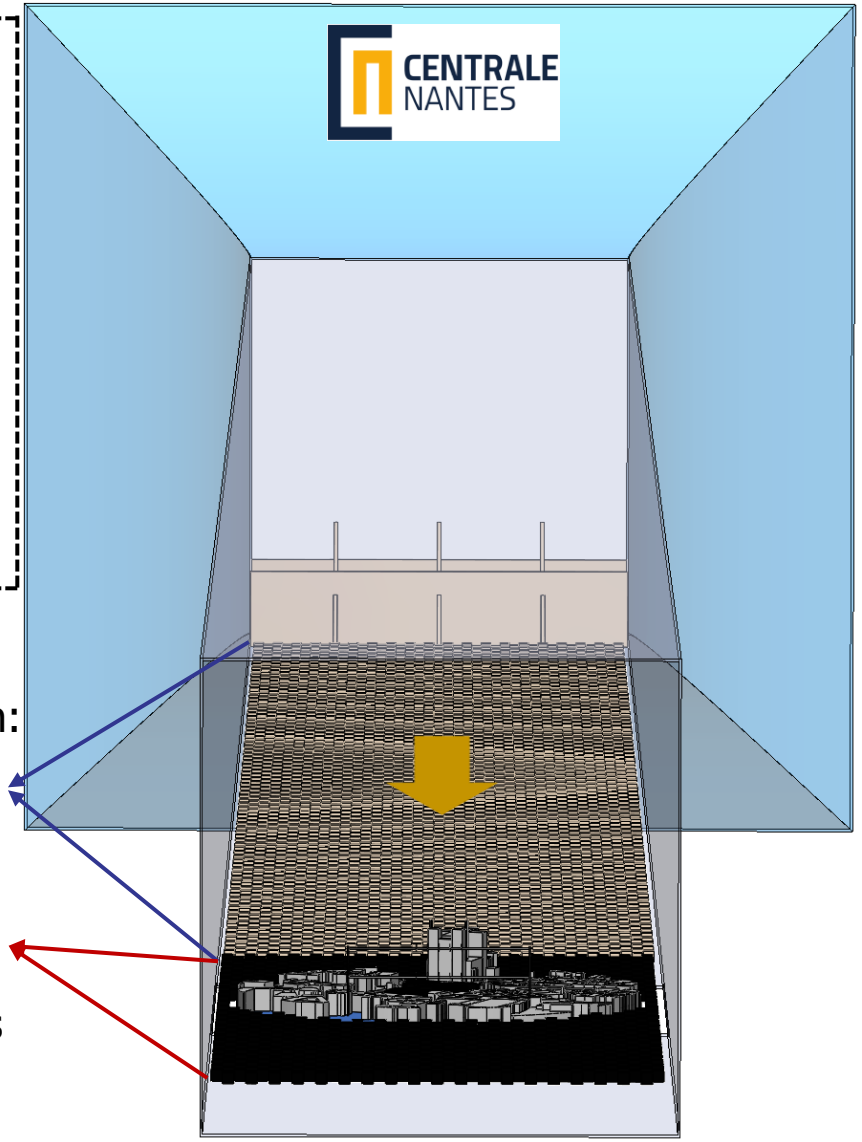
- 1:200 scaled model
- 5 tall building cases:
  - 1/2/3/6 stacked blocks
  - Cathedral
- Block size:

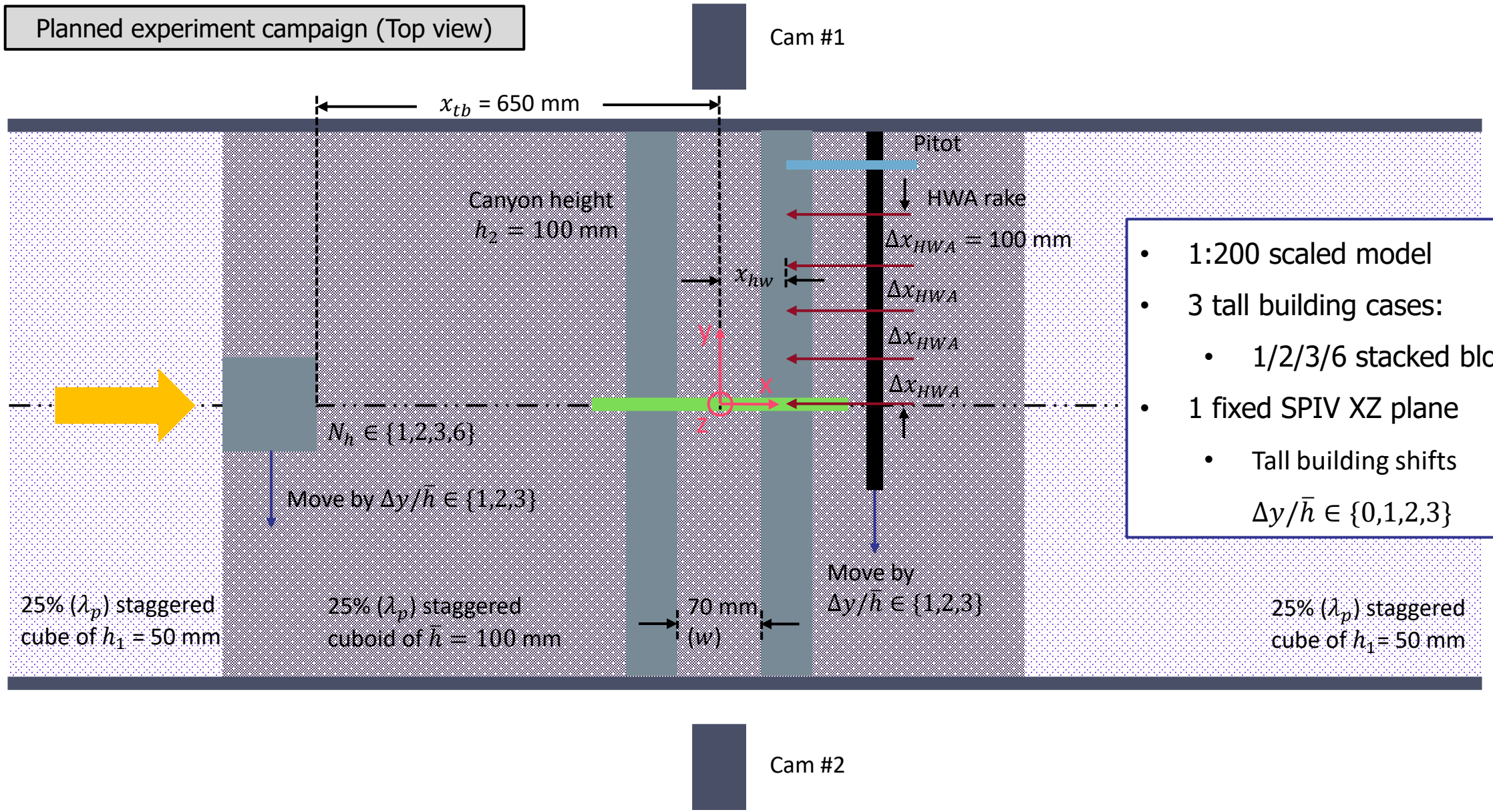
  0.2 m × 0.2 m × 0.1 m
- YZ & XZ planes (separately)

- Ecole Centrale de Nantes
- Dimension before test section:

  18 m (x) × 2 m (y) × 2 m (z)
- Dimension of test section:

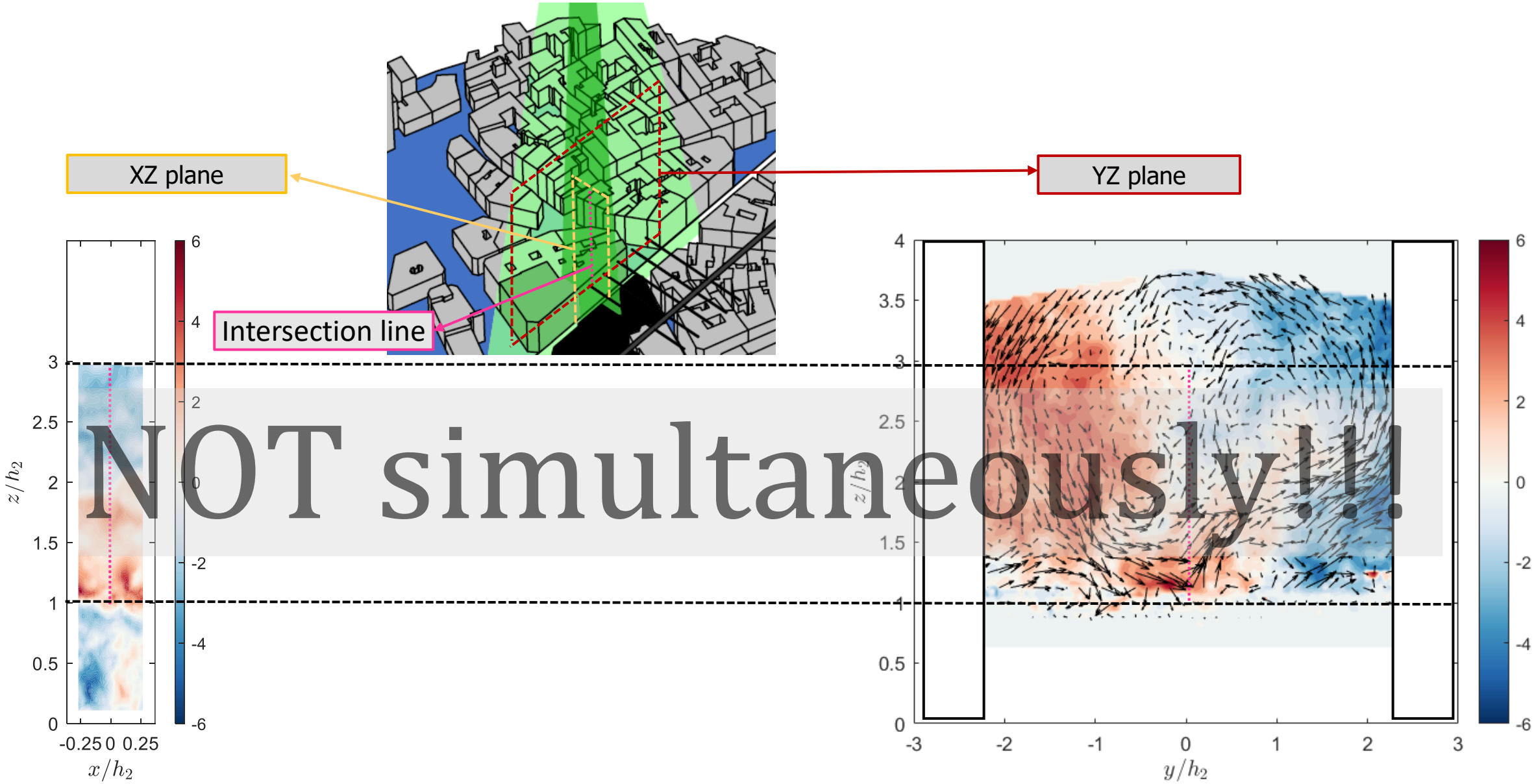
  2 m (x) × 2 m (y) × 2 m (z)
- Free-stream velocity: 5.8 m/s
- Friction velocity: 0.37 m/s



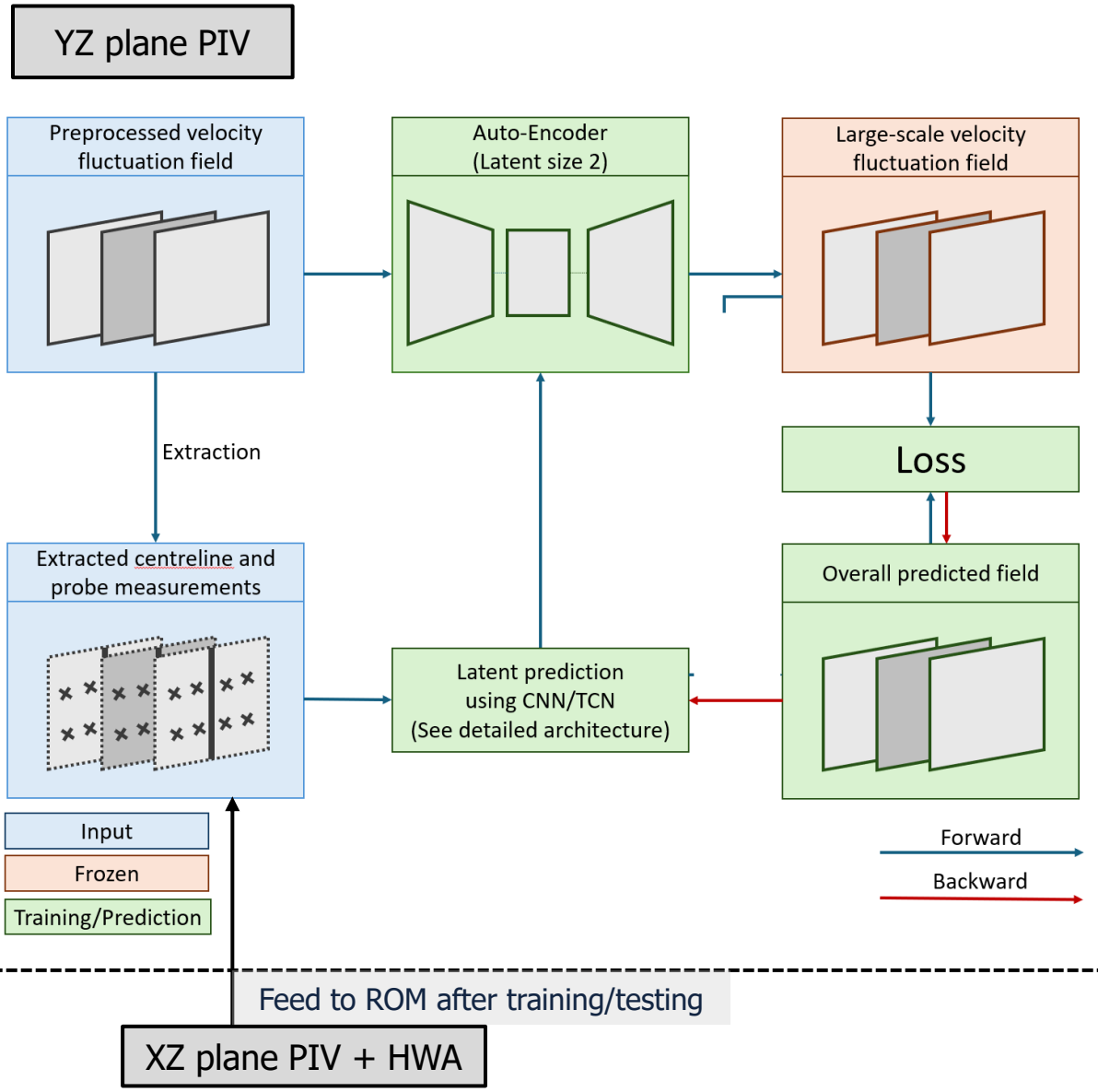


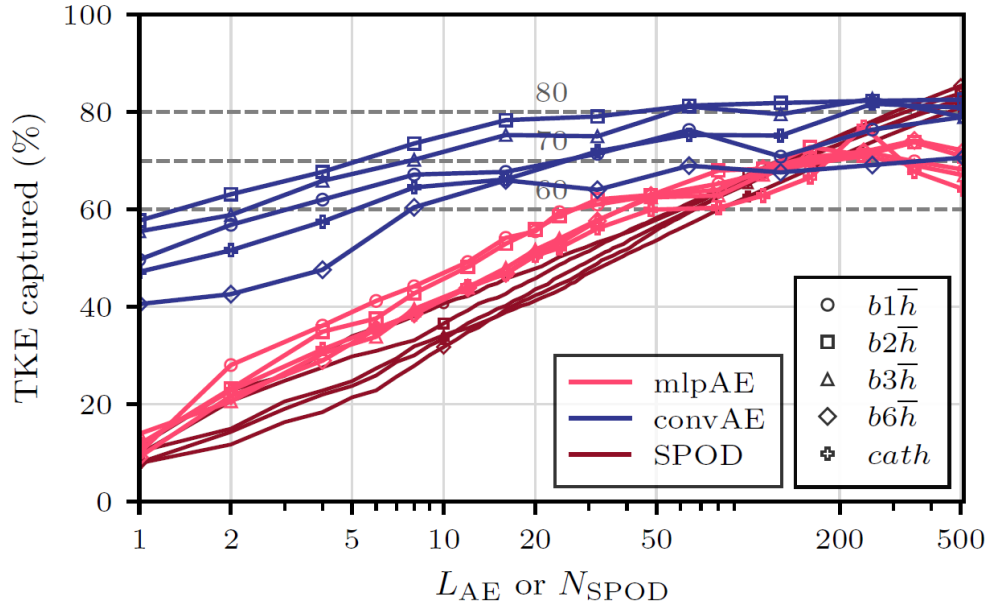
- 1:200 scaled model
- 3 tall building cases:
  - 1/2/3/6 stacked blocks
- 1 fixed SPIV XZ plane
  - Tall building shifts

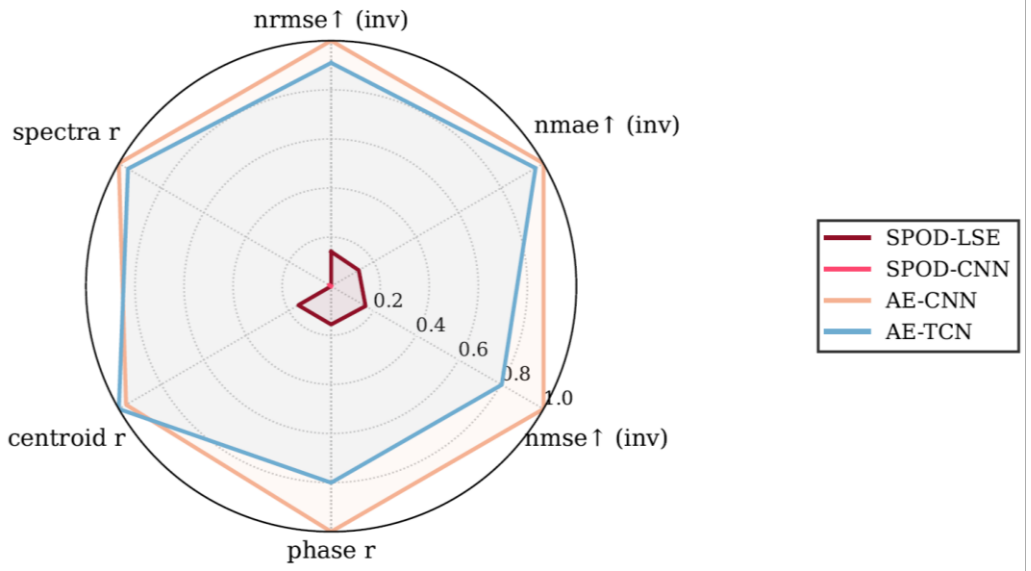
  $\Delta y/\bar{h} \in \{0,1,2,3\}$

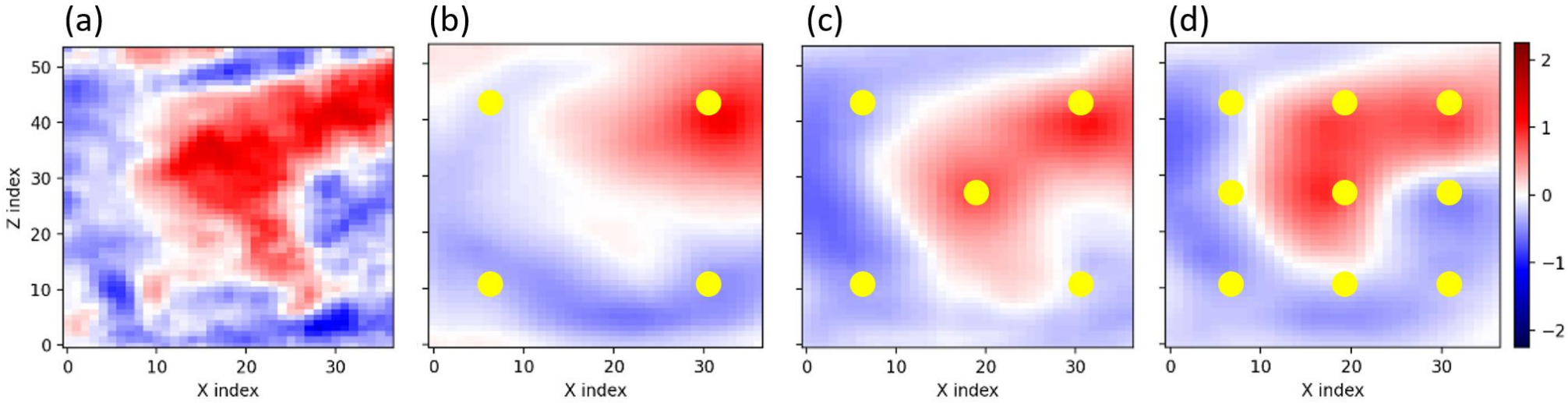

XZ plane
YZ plane
Intersection line
NOT simultaneously!!!
z/h_2
x/h_2
y/h_2
-0.25 0 0.25
-3 -2 -1 0 1 2 3
0 0.5 1 1.5 2 2.5 3 3.5 4
6 4 2 0 -2 -4 -6

> Current Progress > ROM > RdS ABL Reconstruction





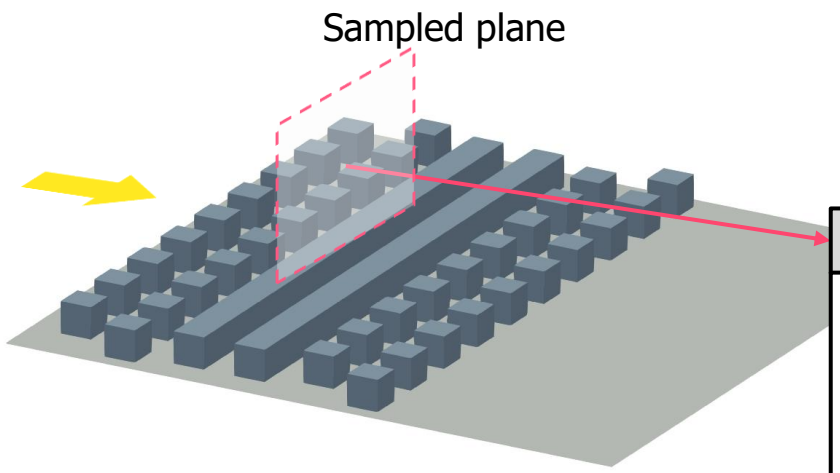




Contours of (a) original transverse velocity fluctuation in RdS dataset, and the reconstructed fields using (b) 4, (c) 5, or (d) 9 synthetic sensors (yellow) at a selected time in RdS dataset using CNN .

Functional baseline 🤔; requires critical updates

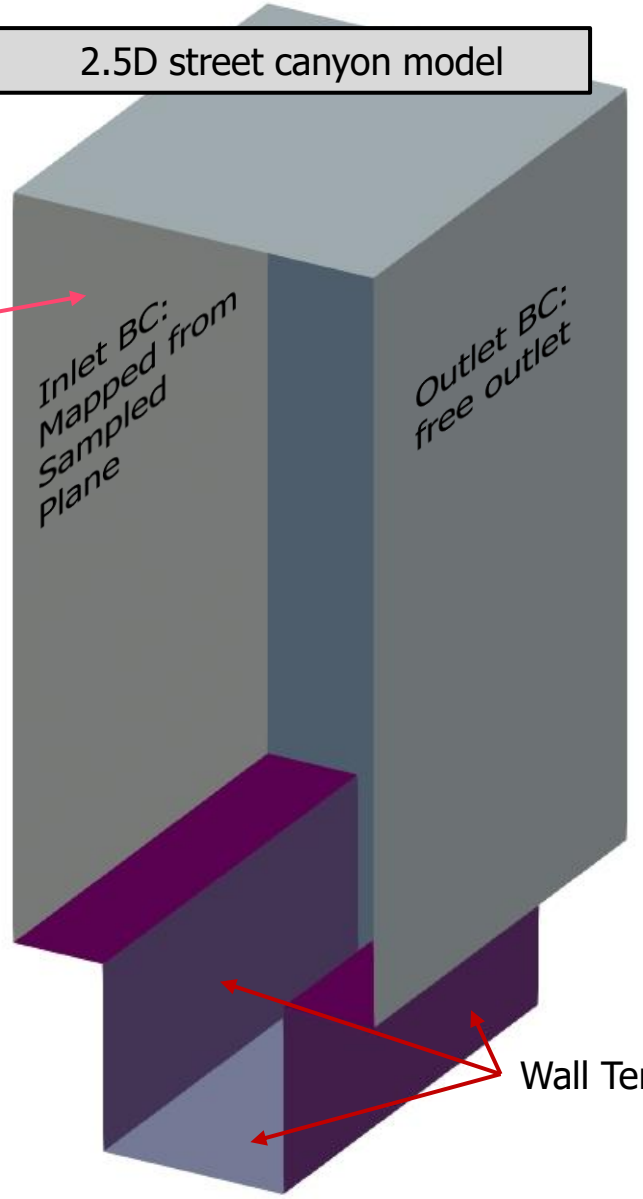
Idealized b1h model (will look slightly different from this model)



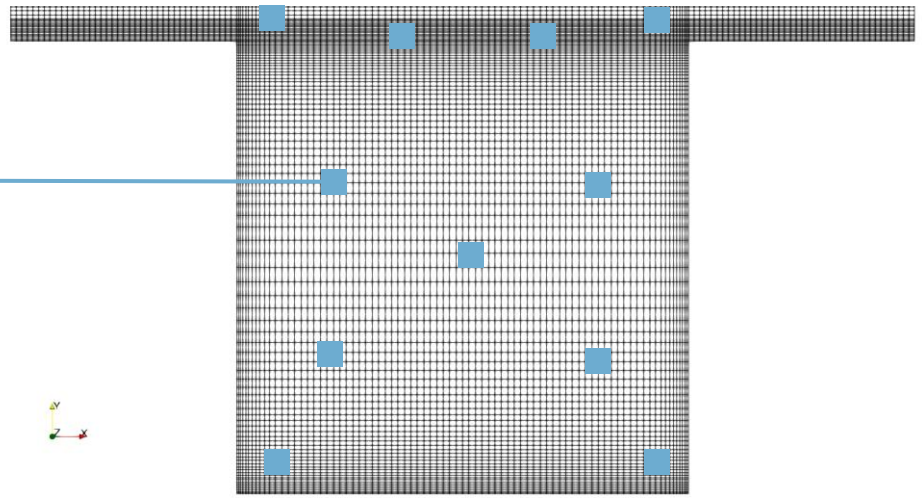
Pre-Rotated Angle

$$U_x' = U_x \cos\phi - U_z \sin\phi,$$
$$U_z' = U_x \sin\phi + U_z \cos\phi,$$
$$U_y' = U_y.$$

2.5D street canyon model



Sensor layout (only for illustration)



Solutions of optimal sensor layout (2D) under inflow angle and heat effects can help with identifying that (3D) for the isothermal morphological case

> Reference

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